MODE STABILITY FOR GRAVITATIONAL PERTURBATIONS OF SUBEXTREMAL KERR–DE SITTER BLACK HOLES

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Abstract. With a computer-assisted proof, we prove mode stability for the Teukolsky equations of spin $s = \pm 2$ on Kerr–de Sitter black hole spacetimes in the full subextremal range: there are no nontrivial outgoing mode solutions at frequencies σ with $\operatorname{Im}\sigma \geq 0$. We then show that every generalized mode solution of the ungauged linearized Einstein equation $D_g\mathrm{Ric}(h) = \Lambda h$ with $\operatorname{Im}\sigma \geq 0$ is pure gauge if $\sigma \neq 0$, and the sum of a linearized Kerr–de Sitter metric and a pure gauge term if $\sigma = 0$. This is the mode stability hypothesis under which Hintz, Petersen and Vasy proved the nonlinear stability of Kerr–de Sitter black holes in the full subextremal range. For real frequencies, the Teukolsky part follows Hintz's two-current method for scalar waves, applied to a dual radial equation obtained from an Euler integral transform. The upper half plane is reached by a continuity argument in which high-frequency control is borrowed from the gauge-fixed Einstein operator through metric reconstruction. The passage from the Teukolsky scalars to the metric uses a linearized Goldberg–Sachs argument and the two Killing fields built from the Killing spinor. Several steps are computer-assisted, using exact arithmetic or floating-point arithmetic with rigorous error bounds.

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1. Introduction

For $\Lambda > 0$, $M > 0$ and $a \in \mathbb{R}$, the Kerr–de Sitter (KdS) metric in Boyer–Lindquist coordinates is

$$g = -\frac{\Delta_r}{\Xi^2 \varrho^2}\big(dt - a\sin^2\theta\, d\varphi\big)^2 + \varrho^2\Big(\frac{dr^2}{\Delta_r} + \frac{d\theta^2}{\Delta_\theta}\Big) + \frac{\Delta_\theta \sin^2\theta}{\Xi^2\varrho^2}\big((r^2+a^2)\, d\varphi - a\, dt\big)^2, \tag{1.1}$$

where

$$\Delta_r = (r^2{+}a^2)\Big(1{-}\frac{\Lambda r^2}{3}\Big){-}2Mr, \quad \varrho^2 = r^2{+}a^2\cos^2\theta, \quad \Delta_\theta = 1{+}\alpha\cos^2\theta, \quad \alpha = \frac{\Lambda a^2}{3}, \quad \Xi = 1{+}\alpha.$$

It solves $\mathrm{Ric}(g) = \Lambda g$. The parameters are *subextremal* if Δ_r has four distinct real roots, which we label

$$r_3 < 0 \le r_0 < r_1 < r_2,$$

where $r_0 = r_C$ is the Cauchy horizon, $r_1 = r_e$ the event horizon and $r_2 = r_c$ the cosmological horizon; then $r_0 + r_1 + r_2 + r_3 = 0$. The domain of outer communications is $r_1 < r < r_2$. We use a time function t_* and an azimuthal angle φ_* that are smooth across the future event and cosmological horizons, as in [5, 4]; a *mode* at frequency $\sigma \in \mathbb{C}$ is a function, or a tensor, of the form $e^{-i\sigma t_*}v$ with v independent of t_* and smooth on $[r_1 - \delta, r_2 + \delta] \times S^2$. In the language of [4], such modes are *outgoing*.

The nonlinear stability of the KdS family was proved by Hintz and Vasy [6] for very slowly rotating black holes. Hintz, Petersen and Vasy [5] extended this to the full subextremal range, conditional on mode stability for the ungauged linearized Einstein equation [5, Assumption 1.2]. Hintz [4] recently proved mode stability for scalar waves in the full subextremal range, and left the Teukolsky equations of spin $s = \pm 2$ open. Our main results address both questions.

Status of this draft. The proof is computer-assisted. Proposition 3.9 (the cosmological side in the near-extremal regime where $L(t) \le 0$) rests partly on verified floating-point computations: Bernstein coefficients are computed exactly, rounded once, and subdivided in double precision with an a priori bound on the accumulated rounding error carried entrywise, and the physical root b_+ is enclosed by interval arithmetic (Appendix A, C9).

Theorem 1.1 (Teukolsky mode stability)**.** *On every subextremal KdS spacetime, the Teukolsky equations of spin $s = +2$ and $s = -2$ have no nontrivial outgoing mode solutions at any frequency $\sigma \in \mathbb{C}$ with $\mathrm{Im}\,\sigma \ge 0$, including $\sigma = 0$. The same holds for generalized modes $\sum_{j<k} t_*^j e^{-i\sigma t_*} v_j$.*

Theorem 1.2 (Ungauged linearized Einstein mode stability)**.** *Let (Λ, M, a) be subextremal with $a \neq 0$, and let $T = \partial_{t_*} + \frac{a}{r_2^2+a^2}\partial_{\varphi_*}$. Let h be a smooth symmetric 2-tensor with $D_g\mathrm{Ric}(h) = \Lambda h$.*

(1) *If* $(\mathcal{L}_T + i\sigma)h = 0$ *with* $\sigma \neq 0$ *and* $\operatorname{Im}\sigma \geq 0$*, then* $h = \mathcal{L}_X g$ *for a smooth vector field* X*.*
(2) *If* $(\mathcal{L}_T)^k h = 0$ *for some* $k \geq 1$*, then* $h = c_M \partial_M g + c_a \partial_a g + \mathcal{L}_X g$ *with* $c_M, c_a \in \mathbb{C}$ *and* X *smooth.*

For $a = 0$, Theorem 1.2 is [6, §7]. Theorem 1.2 is [5, Assumption 1.2] (with $\mathcal{L}_X g = 2\delta^* X^\flat$), so [5, Theorem 1.4] applies (see Remark 1.4 for the choice of T):

Corollary 1.3. *Subextremal Kerr–de Sitter black holes are nonlinearly stable as solutions of the Einstein vacuum equations with positive cosmological constant, in the sense of* [5, Theorem 1.4].

Remark 1.4 (Choice of T). [5] use $T' = \partial_{t_*} + \frac{a}{r_0^2+a^2}\partial_{\varphi_*}$, where r_0 is a critical point of $\mu/(r^2+a^2)$, instead of our T. The two differ by $c\,\partial_{\varphi_*}$ with $c \in \mathbb{R}$, and ∂_{φ_*} commutes with T, T' and $D_g\mathrm{Ric} - \Lambda$. Let h be a (generalized) T'-mode at frequency σ' with $\operatorname{Im}\sigma' \geq 0$. Put it in the gauge of [5, (23)] by adding $\mathcal{L}_Y g$ for a T'-mode Y; this is possible because the gauge equation is solvable on T'-modes [6, §10]. Then h lies in the finite-dimensional space of generalized resonant states of the gauge-fixed operator at σ', which is ∂_{φ_*}-invariant. Hence $h = \sum_{|m|\leq m_0} h_m$ with $\partial_{\varphi_*} h_m = imh_m$, and each h_m is a (generalized) T-mode at frequency $\sigma' - cm$, which has the same imaginary part. Apply Theorem 1.2 to each h_m; for generalized modes with $\sigma \neq 0$ use the reduction in the proof of [5, Theorem 2.7]. If $\sigma' - cm = 0$ and $m \neq 0$, the linearized Kerr–de Sitter part vanishes, because $\partial_M g$ and $\partial_a g$ are axisymmetric. Summing the finitely many pieces shows that Theorem 1.2 implies [5, Assumption 1.2] for T'.

1.1. **Strategy.** The proof of Theorem 1.1 has three parts.

Real frequencies (§§3–4). After separation of variables, outside the superradiant range a Wronskian identity suffices. Inside it, we do not work with the radial Teukolsky equation itself: we apply an Euler (Riemann–Liouville) integral transform to its Heun form. The transform is injective, and its image solves a dual Heun equation that is *real*. Its potential has the very simple form

$$L(z) = \frac{A}{z} + B - 3z, \qquad A > 0,$$

in a variable z in which the event and cosmological horizons sit at $z = 1$ and $z = t$. The dual equation has exactly the structure of the scalar radial equation in [4], so Hintz's two-current method applies. It reduces mode stability to an *event lemma*: an ordinary differential inequality in the four parameters (t, z_0, A, B). A lower angular bound gives the a priori margin $L(z_0) \geq A/(2z_0) + 2t$. We prove the event lemma in three regimes. In the hardest one we follow [4, §4.3.3]: we first saturate one of the two conditions and then solve the other with equality. The remaining claims reduce to sign statements about explicit polynomials, which we certify by exact Bernstein-basis branch and bound.

Zero frequency (§5). This is an energy identity combined with a Teukolsky–Starobinsky map.

The upper half plane (§6). We argue by continuity along a path of subextremal parameters ending at Schwarzschild–de Sitter (SdS), as in [4, §6]. The new difficulty compared with scalar waves is that no high-frequency estimate for the Teukolsky operator in the full subextremal range is available. We circumvent this.

- Using a Teukolsky mode as an outgoing-radiation-gauge Hertz potential [15] produces a linearized Einstein mode with nonvanishing Ψ_4.
- Gauge fixing turns it into a quasinormal mode of the gauge-fixed operator of [5].
- The high-energy estimates of [5] then confine such modes to a compact set.

Nonvanishing of Ψ_4 uses an elementary observation: away from an explicit discrete set of frequencies, nothing annihilated by $(\mathfrak{p}')^4$ can be regular at *both* horizons. This uses the cosmological horizon in an essential way.

Theorem 1.2 (§7). Once the Teukolsky scalars vanish, a linearized Goldberg–Sachs argument shows that the perturbed spacetime is algebraically special to first order. Two Killing fields can then be built covariantly from the perturbed Weyl tensor: the field ξ obtained from the Killing spinor, and Carter's field ζ. Both remain Killing to first order. Moreover, $\operatorname{Im}\delta\xi$ is a Killing field of the background, so it vanishes for every mode with $(\sigma, m) \neq (0, 0)$. This makes h pure gauge, with an explicit gauge vector. The stationary axisymmetric sector reduces to Carter's separable ansatz, on which the field equations are linear.

Computer assistance. Some steps rely on exact symbolic computation or on certificates in exact rational arithmetic. They are listed in Appendix A. The certificates in item C9 use floating-point arithmetic with rigorous a priori error bounds (verified numerics); no step relies on unverified numerical evidence.

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2. Preliminaries

2.1. **Parameters.** By Vieta's formulas,

$$\sum_j r_j = 0, \qquad \sum_{i<j} r_i r_j = -\frac{3(1-\alpha)}{\Lambda}, \qquad r_0 r_1 r_2 r_3 = -\frac{3a^2}{\Lambda}. \tag{2.1}$$

We write $\rho_j = r_j^2 + a^2$, $\Omega_j = a/\rho_j$, $D_j = |\Delta_r'(r_j)|$ and $P_j = \rho_j/D_j$. The signs of $\Delta_r'(r_j)$ are $(-,+,-,+)$ for $j = 0, 1, 2, 3$. Following [4, §6, footnote 5], we use the roots as parameters.

Lemma 2.1 (Rotation bound)**.** *On every subextremal KdS spacetime, $\alpha < 7 - 4\sqrt{3}$.*

Proof. By scaling we may assume $\Lambda = 3$. Then $-\Delta_r = r^4 - (1-\alpha)r^2 + 2Mr - \alpha$, so $p_2 := \sum r_j^2 = 2(1-\alpha)$ and $\mathcal{A} := -r_0 r_1 r_2 r_3 = \alpha$. Scaling a configuration of roots by $\tau > 0$ multiplies $\mathcal{A}$ by τ^4 and p_2 by τ^2. Hence α is the positive root of $\alpha = 1 - \frac{1}{2}\sqrt{\alpha/k}$, where $k := \mathcal{A}/p_2^2$ is scale invariant, and this root is increasing in k. Put $S = -r_3 = r_0 + r_1 + r_2$. By the AM–GM inequality, $r_0 r_1 r_2 \le S^3/27$, and by the QM–AM inequality, $r_0^2 + r_1^2 + r_2^2 \ge S^2/3$. Thus $k \le (S^4/27)/(16S^4/9) = 1/48$, with equality only if $r_0 = r_1 = r_2$, which subextremality excludes. At $k = 1/48$ the root is $\alpha = 12(3^{-1/2} - \frac{1}{2})^2 = 7 - 4\sqrt{3}$. □

2.2. **The Teukolsky equations and separation of variables.** For σ real, the Teukolsky equation of spin s separates as $e^{-i\sigma t + im\varphi} R(r) S(\theta)$ [9]. The angular operator is then symmetric with compact resolvent. As in [4, §3], a mode solution can therefore be decomposed into separated ones, each with an eigenvalue $\lambda \in \mathbb{R}$. We only separate variables for real σ. Put

$$K(r) = \Xi\big((r^2 + a^2)\sigma - am\big), \qquad \kappa_j = \frac{iK(r_j)}{\Delta_r'(r_j)}.$$

The residue theorem applied to ρ/Δ_r, $1/\Delta_r$ and r^k/Δ_r gives

$$\sum_j \kappa_j = 0, \qquad \sum_j r_j \kappa_j = -\frac{3i\Xi\sigma}{\Lambda}, \qquad \sum_j r_j^2 \kappa_j = 0, \tag{2.2}$$

together with $P_1 + P_3 = P_0 + P_2$ and $D_1^{-1} + D_3^{-1} = D_0^{-1} + D_2^{-1}$.

Lemma 2.2 (Boundary conditions)**.** *A separated mode is outgoing if and only if*

$$R \in (r - r_1)^{-\kappa_1 - s}\, C^\infty([r_1, r_2)) \cap (r_2 - r)^{\kappa_2}\, C^\infty((r_1, r_2]).$$

Proof. This follows by writing $t = t_* + T(r)$ and $\varphi = \varphi_* + \Phi(r)$, with the singular parts of T and Φ at r_1 and r_2 as in [4, (3.4)]. The additional factor Δ_r^{-s} at r_1 comes from the boost relating the Kinnersley tetrad to a tetrad that is regular across the event horizon. The Kinnersley tetrad is already regular across the future cosmological horizon in outgoing coordinates. □

2.3. **Heun form.** The Möbius map z with $z(r_0, r_1, r_2, r_3) = (0, 1, t, \infty)$, where

$$t = \frac{(r_1 - r_3)(r_2 - r_0)}{(r_1 - r_0)(r_2 - r_3)} > 1,$$

turns the radial Teukolsky equation into a Heun equation with singular points $0, 1, t, \infty$. We write $\mu(z) = z(z-1)(t-z)$. At z_j ($j = 0, 1, t$) the exponents are those of Lemma 2.2, and at ∞ they are $\{1 + 2s,\ 1 + s - 2\kappa_3\}$. Two constants will appear repeatedly:

$$C_0 = \frac{3}{\Lambda(r_1 - r_0)(r_2 - r_3)}, \qquad c_1 = 1 + t + 3(1 - \alpha)C_0. \tag{2.3}$$

3. Real frequencies: the radial problem

Throughout this section $\sigma \in \mathbb{R} \setminus \{0\}$. We fix a separated outgoing mode with eigenvalue λ. By the symmetries $(\sigma, m, a) \mapsto (-\sigma, -m, a)$ and $(m, a) \mapsto (-m, -a)$ (reflection $\varphi \mapsto -\varphi$) we may assume $m \geq 0$ and $a \geq 0$. For $a = 0$ the window below is empty ($w_1 = w_2 = 0$), so in the window $a > 0$.

3.1. **The superradiant window.** Set $e_1 = 2(\kappa_1 + \kappa_3)$ and $e_t = 2(\kappa_2 + \kappa_3)$. For real σ the κ_j are purely imaginary.

Proposition 3.1 (The window)**.** *Let $m > 0$ and $w = \sigma/m$. Then $\operatorname{Im} e_1 = 2\Xi m f_1(w)$ and $\operatorname{Im} e_t = 2\Xi m f_2(w)$, where*

$$f_1(w) = P_1(w - \Omega_1) + P_3(w - \Omega_3), \qquad f_2(w) = -P_2(w - \Omega_2) + P_3(w - \Omega_3).$$

The function f_1 is increasing and f_2 is decreasing, and their zeros satisfy $0 < w_2 < w_1$. Consequently $\operatorname{Im} e_1 \cdot \operatorname{Im} e_t > 0$ exactly on the open interval (w_2, w_1), the window*.*

Proof. The formulas follow from the definition of κ_j and the signs of $\Delta_r'(r_j)$.

- *Monotonicity.* The slope of f_1 is $P_1 + P_3 > 0$. By the residue identities, the slope of f_2 is $P_3 - P_2 = P_0 - P_1$. Moreover $D_0/D_1 = f(r_0)/f(r_1)$ for the concave parabola $f(x) = (x - r_3)(r_2 - x)$, whose vertex $-(r_0 + r_1)/2$ lies to the left of r_0. Hence $D_0 > D_1$, and with $\rho_1 > \rho_0$ this gives $P_1 > P_0$.
- *Positivity of w_2.* The same comparison gives $w_2 = a(D_1^{-1} - D_0^{-1})/(P_1 - P_0) > 0$.
- *Ordering $w_2 < w_1$.* Cross-multiplying, the terms involving a^2 cancel, and $D_0 D_1 D_3$ times the difference $w_1 - w_2$ factors as $2(r_0 - r_1)(r_0 - r_3)(r_1 - r_3)(r_0 r_3 - r_1 r_2)$. All four factors are negative, since $r_0 r_3 \leq 0 < r_1 r_2$, so the product is positive. □

For every real $\sigma \neq 0$, Proposition 3.4 below puts the dual equation in the real form (3.3), and u is outgoing at both ends. Let B_1, B_t be the boundary coefficients of u at $z = 1, t$ in the sense of [4, (3.5)]. Conservation of $\mu \operatorname{Im}(u'\bar{u})$ gives, as in [4, §3],

$$\operatorname{Im} e_1 \, |B_1|^2 = t \, \operatorname{Im} e_t \, |B_t|^2. \tag{3.1}$$

Corollary 3.2. *There are no outgoing modes with $\sigma \in \mathbb{R} \setminus \{0\}$ outside the open window, nor any with $m = 0$.*

Proof. For $m = 0$, $\operatorname{Im} e_1 = 2\Xi\sigma(P_1 + P_3)$ and $\operatorname{Im} e_t = 2\Xi\sigma(P_3 - P_2) = -2\Xi\sigma(P_1 - P_0)$ have opposite signs. Off the closed window the same holds by Proposition 3.1, so (3.1) forces $B_1 = B_t = 0$. At the endpoints exactly one of $\operatorname{Im} e_1, \operatorname{Im} e_t$ vanishes, since $w_1 \neq w_2$, and (3.1) gives $B_t = 0$ or $B_1 = 0$ at the other end. In each case the indicial roots at that end differ by the non-integer e_t or e_1, so $u = 0$ as after [4, (3.6)]. Then $\tilde{y} = 0$, which contradicts Proposition 3.3. □

3.2. **The dual equation.**

Proposition 3.3 (Euler transform)**.** *Let $\sigma \in \mathbb{R} \setminus \{0\}$, and let $y \neq 0$ be the Heun-form radial function of an outgoing mode, normalized so that it is analytic at $z = t$ and lies in $(z-1)^{\rho_1} C^\omega$ near $z = 1$, where $\rho_1 = -2\kappa_1 - s$. Put $\nu = s - 1 - 2\kappa_3$. Then there is a solution $\tilde{y} \neq 0$ of a Heun equation $\widetilde{\mathcal{H}}\tilde{y} = 0$, obtained from y by a Riemann–Liouville integral of order $\nu + 1$ based at $z = 1$, with the following properties.*

(i) The exponents of $\widetilde{\mathcal{H}}$ at $z_j \in \{0, 1, t\}$ are 0 and $-2(\kappa_j + \kappa_3)$.
(ii) $\tilde{y}$ is analytic at $z = t$, and near $z = 1$ it lies in $(z-1)^{-e_1} C^\omega$, where $e_1 = 2(\kappa_1 + \kappa_3)$.

Proof. That the Euler transform with kernel $(z - w)^\nu$ maps solutions of a Heun equation to solutions of a Heun equation, when $\nu + 2$ is an exponent at ∞, is classical [7, 13]; the exponents in (i) follow from those at ∞ listed in §2. Write $I^{\nu+1}$ for the Riemann–Liouville integral based at $z = 1$, defined on $(z-1)^\beta C^\omega$ by analytic continuation in β; on powers it acts by

$$(z-1)^\beta \mapsto \frac{\Gamma(\beta+1)}{\Gamma(\beta+\nu+2)}(z-1)^{\beta+\nu+1}. \tag{3.2}$$

For $\beta = \rho_1 + k$ we have $\beta + \nu + 2 = k + 1 - e_1$, and $e_1 \in i\mathbb{R}$, so the denominator never sits at a pole of Γ.

The contour. The real integral $\int_1^z (z-w)^\nu y(w)\, dw$ need not converge: $\operatorname{Re}\nu = s - 1$ and $\operatorname{Re}\rho_1 = -s$, so one endpoint is always non-integrable for $s = \pm 2$. We therefore use the Pochhammer double loop $\gamma_z = [1^+, z^+, 1^-, z^-]$ around $w = 1$ and $w = z$, inside a neighborhood of the segment $[1, z]$ on which y is analytic (this is possible for $z \in (1, t]$ near t, since y is analytic on $(1, t]$ and at t). Put

$$\tilde{y}(z) = \oint_{\gamma_z} (z-w)^\nu y(w)\, dw.$$

On powers, $\oint_{\gamma_z} (z-w)^\nu (w-1)^\beta dw = c(\beta)\,(z-1)^{\beta+\nu+1}$ with

$$c(\beta) = \frac{(2\pi i)^2 e^{i\pi(\beta+\nu)}}{\Gamma(-\beta)\Gamma(-\nu)\Gamma(\beta+\nu+2)} \cdot (\pm 1),$$

which is entire in (β, ν). It vanishes only if $\beta \in \mathbb{Z}_{\geq 0}$, or $\nu \in \mathbb{Z}_{\geq 0}$, or $\beta + \nu + 2 \in \mathbb{Z}_{\leq 0}$. When the real integral converges, $c(\beta)$ is a nonzero multiple of (3.2). In the window $\kappa_3 \notin \mathbb{R}$, so $\nu \notin \mathbb{Z}$. For $\beta = \rho_1 + k$ we have $\beta + \nu + 2 = k + 1 - e_1$ with $e_1 \in i\mathbb{R}$, so the last condition fails for $k \geq 0$.

Generic case $\kappa_1 \neq 0$. Then $\rho_1 \notin \mathbb{R}$, so $c(\rho_1 + k) \neq 0$ for all $k \geq 0$. Termwise integration of $y = (z-1)^{\rho_1} \sum_k a_k (z-1)^k$ gives $\tilde{y} = (z-1)^{-e_1} \sum_k c(\rho_1 + k) a_k (z-1)^k$, which is nonzero.

Threshold case $\kappa_1 = 0$. Then $\rho_1 = -s$ is an integer, and $c(\beta)$ vanishes on the analytic part ($\beta \in \mathbb{Z}_{\geq 0}$). For $s = -2$, $\rho_1 = 2$, so y is analytic with leading power $(z-1)^2$. Use the Hankel-type contour from the side, i.e. $\tilde{y} = \Gamma(-\nu)^{-1}$-normalised $\int_1^z$, which converges at $w = 1$; the divergence at $w = z$ ($\operatorname{Re}\nu = -3$) is removed by the loop around $w = z$ alone. On powers this gives $\Gamma(\beta+1)/\Gamma(\beta+\nu+2)$ up to a nonzero factor, and this is nonzero for $\beta = 2 + k$. For $s = +2$, $\rho_1 = -2$: write $y = (z-1)^{-2}(a_0 + a_1(z-1)) + y_{\mathrm{an}}$. If $(a_0, a_1) \neq 0$, the Pochhammer transform kills y_{an} and maps $(z-1)^{-2}, (z-1)^{-1}$ to nonzero multiples of $(z-1)^{\nu-1}, (z-1)^\nu$, since $c(-2), c(-1) \neq 0$. If $a_0 = a_1 = 0$, y is analytic and we use the loop around $w = z$ alone, as for $s = -2$. In all cases $\tilde{y} \neq 0$ and $\tilde{y} \in (z-1)^{-e_1} C^\omega$.

Analyticity at $z = t$ and the equation. The integrand is analytic on the contour and depends analytically on z, and the contour can be deformed locally as z moves, so $\tilde{y}$ is analytic near $z = t$. The Euler transform intertwines the two Heun operators up to an exact derivative $\partial_w[(\cdots)(z-w)^\nu \cdots]$; this integrates to zero over a closed (Pochhammer) contour, and for a loop around $w = z$ alone the integrand returns to its starting value. So $\widetilde{\mathcal{H}}\tilde{y} = 0$. □

Proposition 3.4 (The real dual)**.** *For real $\sigma \neq 0$, after multiplication by an explicit nonvanishing gauge factor, the equation $\widetilde{\mathcal{H}}\tilde{y} = 0$ becomes*

$$-(\mu u')' - \frac{q(z)^2}{\mu}u + L(z)u = 0, \qquad q(z) = q_1(z - z_0), \qquad L(z) = \frac{A}{z} + B - 3z, \tag{3.3}$$

with real coefficients, q affine, and $q(1)$, $q(t)$ proportional (with a common sign) to $\frac{1}{2}(t-1)\operatorname{Im} e_1$ and $-\frac{1}{2}t(t-1)\operatorname{Im} e_t$. If w lies in the open window, then moreover:

- *$q = q_1(z - z_0)$ with $q_1 > 0$ and $z_0 \in (1, t)$, after replacing u by $\bar{u}$ and q by $-q$ if necessary;*
- *$A = t\big(X_0^2 - (X_1 + X_t)^2\big)$ with $X_j = |\kappa_j + \kappa_3|$, and $A > 0$;*
- *$B = \lambda C_0 + c_1$.*

The transformed solution u is outgoing for (3.3) in the sense of [4, (3.4)]: near $z = 1$ it is $(z-1)^{-iq(1)/\mu'(1)}$ times a smooth function, and near $z = t$ it is $(t-z)^{-iq(t)/|\mu'(t)|}$ times a smooth function.

Proof. For real σ all exponents of $\widetilde{\mathcal{H}}$ at $0, 1, t$ are purely imaginary, and the gauge factor $\prod_j (z - z_j)^{-\operatorname{Re}(\cdot)}$ removes the real parts. The resulting operator is formally real, and writing it in the form (3.3) determines $q(1)$ and $q(t)$ from the exponents at $z = 1$ and $z = t$. We take q to be linear. By Proposition 3.1, $q(1)q(t) < 0$, so q has a unique zero $z_0 \in (1, t)$, and the sign of q_1 is fixed by replacing u with $\bar{u}$ if necessary. The remaining claims are as follows.

- *Form of L.* The function $L := (q^2 - \mu^2 V)/\mu$ is then rational, and $L = n_0/z + n_1 + n_2 z$. Expanding at $z = \infty$ with the exponents $\{1 + 2s, 1 + s - 2\kappa_3\}$ gives $n_2 = -3$, since the exponent difference there is ± 4.
- *The coefficient n_0.* Expanding at $z = 0$ gives $n_0 = t(X_0^2 - (X_1 + X_t)^2)$. In the window $X_0 - X_1 - X_t = 2\operatorname{Im}\kappa_3$, so $n_0 = 2t\,\operatorname{Im}\kappa_3\,(X_0 + X_1 + X_t)$. This is positive because $\operatorname{Im}\kappa_3 > 0$ for $w > \Omega_3$, and $\Omega_3 < \Omega_2 < w$ since $|r_3| > r_2$.
- *The coefficient n_1.* The identity $n_1 = \lambda C_0 + c_1$ is an exact identity between rational functions of $(r_0, r_1, r_2, \sigma, m, \lambda, a, s)$ after elimination by (2.1) and (2.2). Its verification is computer algebra (Appendix A, item C1): the numerators of $n_1 - (\lambda C_0 + c_1)$ and $n_2 + 3$ are divisible by $s^2 - 4$.
- *Outgoing behavior.* This follows from Proposition 3.3(ii) and the matching of exponents. □

Lemma 3.5 (A priori bounds)**.** *Throughout the open window, $L' < 0$ on $[1, t]$, and for every $m \geq 1$*

$$L(z_0) \;\geq\; \frac{A}{2z_0} + 2t. \tag{3.4}$$

Proof. First, $L' = -A/z^2 - 3 < 0$. For (3.4) we use only a crude lower bound for the angular eigenvalue, $\lambda \geq \lambda_{\mathrm{cr}} := \ell_0(\ell_0+1) - 4 - 4\alpha - 4\alpha|m|(1+\alpha) - 2(1+\alpha)^2|\xi|(|m|+2)$ with $\ell_0 = \max(|m|, 2)$ (Proposition 3.6). Since $L(z_0)$ is increasing in $B = \lambda C_0 + c_1$, it suffices to prove (3.4) with $\lambda = \lambda_{\mathrm{cr}}$. Then $\Lambda/3$ times $L(z_0) - A/(2z_0) - 2t$ is a quadratic polynomial $m^2 Q_2 + mQ_1 + Q_0$ (with the modification $+4C_0\Lambda/3$ at $m = 1$) whose coefficients are rational functions of (r_0, r_2, a^2, τ), where $\tau \in (0, 1)$ parametrizes the window. One has $Q_0 < 0$, so it suffices to show positivity at $m = 1$, at $m = 2$ and of Q_2. After eliminating a^2 by the quadratic relation it satisfies, each of these

reduces to the positivity of finitely many polynomials on the open cube in $(r_0, 1/r_2, \tau)$. All of them have nonnegative Bernstein coefficients on $[0,1]^3$, not all zero, which proves positivity on the open cube (Appendix A, item C2). □

Proposition 3.6 (Crude angular bound)**.** *For $s = \pm 2$, $m \geq 0$, $\alpha \geq 0$ and $\xi = a\sigma \geq 0$ (in particular on the window), $\lambda \geq \lambda_{\mathrm{cr}}$.*

Proof. In the variable $x = \cos\theta$, the angular operator of [12, (2.3)] is $H = H_0 + K_\alpha + V$, where

$$H_0 = -\partial_x(1-x^2)\partial_x + \frac{(m+sx)^2}{1-x^2}, \qquad K_\alpha = -\partial_x\, \alpha x^2(1-x^2)\,\partial_x \;\geq 0,$$

and λ is the lowest eigenvalue of H in the sector (m, s). On that sector $H_0 \geq \ell_0(\ell_0+1) - s^2$ with $\ell_0 = \max(|m|, |s|)$. An exact computation gives $V = \frac{\alpha(m+s)^2}{2(1-x)} + \frac{\alpha(m-s)^2}{2(1+x)} + \frac{\mathcal{N}}{1+\alpha x^2}$, where the first two terms are nonnegative and

$$\mathcal{N} = \alpha^2 m^2 - 2\alpha^2 msx + 2\alpha^2 x^4 + \alpha m^2 - 2\alpha msx + \alpha s^2 x^2 - \alpha s^2 + 2\alpha x^2 + \xi^2(1+\alpha)^2(1-x^2) - 2\xi(1+\alpha)^2(m-sx).$$

Let $T = -4\alpha - 4\alpha m(1+\alpha) - 2(1+\alpha)^2\xi(m+2)$. With $s^2 = 4$,

$$\begin{aligned}\mathcal{N} - T(1+\alpha x^2) = {} & (2\alpha^2 x^4 + 4\alpha^2 x^2 + 6\alpha x^2) + (\alpha^2+\alpha)m^2 \\ & + m\big(4\alpha^3 x^2 + 4\alpha^2 x^2 + 4(\alpha^2+\alpha)(1 - \tfrac{s}{2}x)\big) + \xi^2(1+\alpha)^2(1-x^2) \\ & + \xi\big(2\alpha^3 m x^2 + 4\alpha^3 x^2 + 4\alpha^2 m x^2 + 8\alpha^2 x^2 + 2\alpha m x^2 + 4\alpha x^2 + 2(1+\alpha)^2(2+sx)\big).\end{aligned}$$

Every term is nonnegative for $|x| \leq 1$, $|s| = 2$, $m, \alpha, \xi \geq 0$. Hence $V \geq T$, and $\lambda \geq \ell_0(\ell_0+1) - 4 + T = \lambda_{\mathrm{cr}}$. □

Remark. An earlier version of this paper claimed $\lambda \geq \frac{3}{2}$. That claim was based on a sign error in the angular operator and is false: near extremality, λ can be negative for $m = 2$. Nothing below uses it.

3.3. **Currents.** On the dual equation (3.3) we use the currents of [4, Lemma 4.1], in the form with variable potential from [4, §7]. Let $u_\pm = \exp(\mp i \int_{z_0}^{z} q/\mu)\, u$; these are smooth down to $z = t$ and $z = 1$, respectively. For a real function f, define

$$\begin{aligned} J^f[v] &= -\mu f q|v'|^2 - \mu f q_1 \operatorname{Re}(v'\bar{v}) + \big(Lfq + \tfrac{1}{2}q_1\mu f'\big)|v|^2, \\ C^f &= \mu' f q - 2\mu f q_1 - \mu f' q, \qquad D^f = q(fL)' + \tfrac{1}{2}q_1(\mu f')'. \end{aligned}$$

Then $\frac{d}{dz}J^f[u_\pm] = C^f|u_\pm'|^2 + D^f|u_\pm|^2$. The proof is the integration by parts of [4, Lemma 4.1], using $q' = q_1$.

Lemma 3.7 (Horizon inequality)**.** *For all $z_0 \in [1,t]$ and $z \in (1,t)$, we have $H(z) := 2\mu(z) + (z_0 - z)\mu'(z) > 0$.*

Proof. H is affine in z_0. At $z_0 = 1$ it equals $(z-1)\big((z-1)^2 + t - 1\big) > 0$. At $z_0 = t$ it equals $(t-z)\big(t(t-1) - (t-z)^2\big) > 0$. □

Proposition 3.8 (Reduction to the event lemma)**.** *Suppose there is a function $f \in C^2((1,z_0]) \cap C^0([1,z_0])$ with the following properties:*

(a) $f(1) = 0$, $f'(z) = o\big((z-1)^{-1}\big)$ as $z \to 1$, and $f > 0$ on $(1, z_0]$;
(b) $f'(z_0) \geq 0$;
(c) with $s := z_0 - z$ and $h := f'/f$, on $(1, z_0)$,

$$(C) \quad s\,h\,\mu \leq H, \qquad\qquad (D) \quad \tfrac{1}{2}\big(\mu' h + \mu(h' + h^2)\big) \leq s\,(Lh + L'). \tag{3.5}$$

Then every outgoing solution of (3.3) vanishes.

Proof. Cosmological side. If $L(t) > 0$, take $f_+ = 1$ on $[z_0, t]$. Then $C^{f_+} = -q_1 H \le 0$ by Lemma 3.7, and $D^{f_+} = qL' \le 0$ because $q \ge 0$ there. The limits of the current are computed as in [4, Lemma 4.2]:

$$\lim_{z\to t} J^{f_+}[u_+] = L(t)q(t)|B_t|^2, \qquad J^{f_+}[u_+](z_0) = -\mu(z_0)q_1 \operatorname{Re}(u'\bar{u})(z_0).$$

Since J^{f_+} is non-increasing and $L(t)q(t) > 0$, we get $\operatorname{Re}(u'\bar{u})(z_0) \le 0$, with equality only if $B_t = 0$. If $L(t) \le 0$, the same argument is run with the bent phase $\tilde{q} = (z - z_0)\big(q_1 + \kappa(t - z)\big)$ and the correspondingly modified $\tilde{L} = L + (\tilde{q}^2 - q^2)/\mu$, for a $\kappa > 0$ with $\tilde{L}(t) > 0$; see Proposition 3.9 below.

Event side. On $[1, z_0]$, write $q = -q_1 s$. Then $C^f \le 0$ is equivalent to (C), and $D^f \le 0$ is equivalent to $(\mu f')' \le 2s(Lf)'$, which is (D). By (a), $\lim_{z\to 1} J^f[u_-] = L(1)f(1)q(1)|B_1|^2 = 0$. Monotonicity gives

$$J^f[u_-](z_0) = -\mu(z_0)q_1 f(z_0) \operatorname{Re}(u'\bar{u})(z_0) + \tfrac{1}{2}q_1\mu(z_0)f'(z_0)|u(z_0)|^2 \le 0.$$

So $\operatorname{Re}(u'\bar{u})(z_0) \ge 0$ by (b).

Conclusion. Comparing the two sides forces $B_t = 0$. At $z = t$ the indicial roots differ by a non-integer, because $\operatorname{Im} e_t \ne 0$ in the open window, so the regular-singular argument following [4, (3.6)] gives $u = 0$. □

By Propositions 3.3, 3.4 and 3.8, the real-frequency part of Theorem 1.1 follows from the next result, which only involves the four numbers (t, z_0, A, B).

Proposition 3.9 (Cosmological side for $L(t) \le 0$)**.** *Let w lie in the open window and suppose $L(t) \le 0$. Then there is $\kappa > 0$ such that the bent phase $\tilde{q} = (z - z_0)\big(q_1 + \kappa(t - z)\big)$ and $\tilde{L} = L + (\tilde{q}^2 - q^2)/\mu$ satisfy $\tilde{L}(t) > 0$, $\tilde{C} := \mu'\tilde{q} - 2\mu\tilde{q}' \le 0$ and $\tilde{D} := \tilde{q}\tilde{L}' - \kappa\mu' \le 0$ on (z_0, t).*

With $f_+ = 1$, the current argument of [4, Lemma 4.1, §7] applies verbatim to the phase $\tilde{q}$: the derivative of the current is $\tilde{C}|u'_+|^2 + \tilde{D}|u_+|^2$, where $\tilde{D} = \tilde{q}\tilde{L}' + \frac{1}{2}(\mu\tilde{q}'')'$ and $\tilde{q}'' = -2\kappa$. Since $\tilde{q}(t) = q(t)$, the boundary limit at t is $\tilde{L}(t)q(t)|B_t|^2$. At z_0 the current equals $-\mu(z_0)\tilde{q}'(z_0)\operatorname{Re}(u'\bar{u})(z_0)$ with $\tilde{q}'(z_0) = q_1 + \kappa(t - z_0) > 0$. So Proposition 3.9 gives the cosmological half of Proposition 3.8 when $L(t) \le 0$.

Proof. Write $\delta = z_0 - 1 > 0$ and $\theta = t - z_0 > 0$. Note $\tilde{L}(t) = L(t) + 2\kappa q_1\theta^2/(t(t-1))$.

Step 0 (the crude eigenvalue suffices). Let L_{cr} be L with λ replaced by $\lambda_{\mathrm{cr}} \le \lambda$ (Proposition 3.6). Then $L \ge L_{\mathrm{cr}}$, while q_1, z_0, A and the expressions $\tilde{C}, \tilde{D}$ do not involve λ ($\tilde{L}' = L' + ((\tilde{q}^2 - q^2)/\mu)'$ and $L' = -A/z^2 - 3$). Hence it suffices to find $\kappa > 0$ with $\tilde{L}_{\mathrm{cr}}(t) > 0$, $\tilde{C} \le 0$ and $\tilde{D} \le 0$. In what follows $L(t)$ means $L_{\mathrm{cr}}(t)$, and we may assume $L(t) \le 0$. If $L(t) < 0$ we take $\kappa = 1.05\,\kappa_{\min}$ with $\kappa_{\min} = -L(t)t(t-1)/(2q_1\theta^2)$. If $L(t) = 0$ we take $\kappa > 0$ small; every condition below is then satisfied for small κ by inspection, given (i) of Step 6.

Step 1 ($\tilde{C}$). One computes $\tilde{C} = -q_1 H - \kappa(t - z)F$, with H as in Lemma 3.7 and $F = z_0(2tz - t - z^2) - tz - z^3 + 2z^2$. Substituting $z_0 = 1 + \delta$, $z = z_0 + y$, $t = z + c$ ($\delta, y, c > 0$), the polynomial $\theta H + (t - z)F$ has only positive coefficients, so $(t - z)(-F) \le \theta H$. Hence $\tilde{C} \le -H(q_1 - \kappa\theta) \le 0$ as soon as $k\theta \le 1$, where $k := \kappa/q_1$. Write $\varpi := k\theta$.

Step 2 ($\tilde{D}$). Multiply $\tilde{D}$ by $q_1^{-3}z^2(z-1)^2 > 0$ and expand in (δ, y, c) (an exact polynomial identity). Apart from $-A\,y(z-1)^2(1+ck)$ and $-3yz^2(z-1)^2$, every term carries the factor k. Grouping, $\tilde{D} \le 0$ on (z_0, t) is equivalent to $A \ge P_1(y) + q_1^2 P_2(y)$ for $y \in (0, \theta)$, where, with $c = \theta - y$,

$$P_1 = \frac{z^2(gk - 3y)}{y(1 + ck)}, \quad g = y(2{+}4\delta{+}y{-}5c){-}M_0, \quad M_0 = \theta(1{+}2\delta){-}\delta(1{+}\delta), \qquad P_2 = \frac{yk\big[(2\delta z + y)(2 + ck) - kyz(z$$

Note $M_0 = \mu'(z_0)$. If $M_0 \geq 0$ then $g \leq y(2+4\delta+\theta)$, so $P_1 \leq 0$ provided

$$(S_\alpha) \quad k(2+4\delta+\theta) \leq 3.$$

So $\tilde{D} \leq 0$ follows from (S_μ): $M_0 \geq 0$, together with (S_α) and (S_β): $A \geq q_1^2 \sup_{0<y<\theta} P_2(y)$.

Step 3 (z_0-free form of ϖ). By Proposition 3.1, $\operatorname{Im} e_1 < 0$ and $\operatorname{Im} e_t < 0$ in the window. Matching exponents in Proposition 3.4 gives $q_1 = (t-1)|\operatorname{Im} e_1|/(2\delta)$ and $z_0 = 1 + (t-1)\,|\operatorname{Im} e_1|/(|\operatorname{Im} e_1| + t|\operatorname{Im} e_t|)$. Substituting into $\varpi = 1.05\,(-L(t))t(t-1)/(2q_1^2\theta)$ gives the exact identity

$$\varpi = \frac{2.1\,(-L(t))}{|\operatorname{Im} e_t|\,\big(|\operatorname{Im} e_1| + t|\operatorname{Im} e_t|\big)}.$$

Write $\tilde{\lambda} = \Lambda/3$ and use $\operatorname{Im} e_1 = 2\Xi a m f_1(w)/\tilde{\lambda}$, $\operatorname{Im} e_t = 2\Xi a m f_2(w)/\tilde{\lambda}$ (Proposition 3.1, with Λ restored), where here a is the rotation parameter. Then $\varpi = \varpi_m := 2.1\,(-H_m)\tilde{\lambda}/\big(4\Xi^2 a^2 f_2(f_1 + tf_2)\big)$ with $H_m := \tilde{\lambda}L(t)/m^2$. The denominator is positive and independent of m. Moreover $H_m = \ell_2 + \ell_1 u + \ell_0 u^2$ with $u = 1/m$, where $\ell_0 = -K(1+7\alpha) + \tilde{\lambda}(1-2t) < 0$ ($K > 0$ explicit) and $\ell_2 > 0$ on the whole window (certificate, item C9). So ϖ_m is convex in u, and $\varpi_m \leq \max(\varpi_2, \varpi_\infty) \leq \max(\varpi_2, 0)$ for $m \geq 2$. For $m = 1$ the m-dependence of λ_{cr} differs, and ϖ_1 is treated separately. Hence $\varpi \leq \bar{\varpi} := \max(\varpi_1, \varpi_2, 0)$.

Step 4 (smooth variables). Put $\gamma = A/q_1^2$, $\eta = \delta/\theta$ and $\iota = 1/\theta$. These, and ϖ_m, are rational in the window variables, independent of m (except ϖ_m), and extend smoothly to the extremal face $r_0 = r_1$. Then $(S_\alpha) \iff \varpi(1+4\eta+2\iota) \leq 3$. With $y = \theta v$, $v \in (0,1)$, and using $z - 1 = \theta(\eta + v)$ and $z = \theta(\iota + \eta + v)$, (S_β) becomes: for all $v \in (0,1)$,

$$W(v) := \gamma(\eta+v)^2 - 2\varpi v P - \varpi^2 v E \geq 0, \qquad P = 2\eta(\eta+v) + (2\eta+v)\iota, \quad E = P(1-v) - v(\eta+v)(\eta+v+\iota).$$

Step 5 (eliminating v). Let $\varpi > 0$. Since $E \leq P(1-v)$ and $P \leq 2(\eta+v)(\eta+\iota)$,

$$W(v) \geq (\eta+v)\big[\gamma(\eta+v) - (4\varpi + 2\varpi^2)v(\eta+\iota)\big] = (\eta+v)\big[(1-v)\gamma\eta + v\,Z(\varpi)\big], \qquad Z(\varpi) := \gamma(1+\eta) - (4\varpi+2\varpi^2)(\eta+\iota).$$

The bracket is affine in v, so $W > 0$ on $(0,1)$ if $Z(\varpi) > 0$ (recall $\gamma\eta > 0$). Both Z and $\varpi \mapsto 3 - \varpi(1+4\eta+2\iota)$ are decreasing for $\varpi \geq 0$, and $\varpi \leq \bar{\varpi}$. So (S_α) and (S_β) follow from: for $m = 1, 2$, either $\varpi_m \leq 0$, or $Z(\varpi_m) > 0$ and $3 - \varpi_m(1+4\eta+2\iota) > 0$.

Step 6 (certificates). The following are proved on the whole window by Bernstein subdivision in the variables $(r_0, 1/r_2, \tau)$, with blow-up charts at the extremal, Kerr, Nariai and ultracold faces (Appendix A, item C9): (i) $L(t) \leq 0 \Rightarrow z_f := \delta/(t-1) < \frac{1}{2}$; (ii) $\varpi_m \leq 1$ for $m = 1, 2$; (iii) for $m = 1, 2$: $\varpi_m \leq 0$ or $3 - \varpi_m(1+4\eta+2\iota) > 0$; (iv) for $m = 1, 2$: $\varpi_m \leq 0$ or $Z(\varpi_m) > 0$; (v) $f_1 < 0$, $f_2 < 0$ and $\ell_2 > 0$. From (i), $z_0 < (1+t)/2$; since μ' is a concave quadratic with $\mu'(1) = t - 1 > 0$ and $\mu'((1+t)/2) = (t-1)^2/4 > 0$, we get $M_0 = \mu'(z_0) > 0$. By Steps 1–5, (i)–(v) give $\tilde{C} \leq 0$, $\tilde{D} \leq 0$ and $\tilde{L}(t) > 0$. □

The physical margins are ≥ 5.2 for (i), ≥ 1.55 for (ii), ≥ 4.66 for (iii) and ≥ 1.077 for (iv); the last is the thinnest and is attained in the extremal limit near the top of the window.

Theorem 3.10 (Event lemma). *For all $t > 1$, $z_0 \in (1, t)$ and $A > 0$ with $L(z_0) \geq A/(2z_0) + 2t$, there exists f satisfying (a)–(c) of Proposition 3.8.*

4. Proof of the event lemma

In terms of f, condition (C) reads $f'/f \leq h_C := H/(s\mu) = 2/s + \mu'/\mu$. Condition (D) reads $\widetilde{D}f := (\mu f')' - 2s(Lf)' \leq 0$. Put $x_0 = z_0 - 1$ and

$$p = \frac{2x_0 L(1)}{t-1}.$$

We distinguish three regimes:

$$\text{R1: } p \le 1; \qquad \text{R2: } p > 1 \text{ and } 2x_0^2\Big(\frac{A}{z_0}+3\Big) < t-1; \qquad \text{R3: the rest.}$$

4.1. **Regimes R1 and R2: an explicit solution.**

Proposition 4.1. *In R1 and R2, the conclusion of Theorem 3.10 holds.*

Proof. Let $\tilde{L} = L$ in R1, and $\tilde{L} = L - \kappa$ with $\kappa = L(1) - \frac{t-1}{2x_0} > 0$ in R2. Let f solve $(\mu f')' = 2s(\tilde{L}f)'$ with $f(1) = 0$, that is,

$$\mu f' = 2s\tilde{L}f + 2\int_1^z \tilde{L}f.$$

At $z = 1$ this is a regular-singular equation, and $f \sim (z-1)^{\tilde{p}}$ with $\tilde{p} = 2x_0\tilde{L}(1)/(t-1) \in (0,1]$. So (a) holds. In R2 we have $\tilde{L}(z_0) > 0$, since $L(1) - L(z_0) = Ax_0/z_0 + 3x_0$ and this is exactly the R2 condition. Thus $\tilde{L} > 0$ on $[1, z_0]$, so $\mu f' > 0$ and f is positive and increasing; in particular (b) holds.

For (D): $(\mu f')' = 2s(\tilde{L}f)' \le 2s(Lf)'$, because $((L-\tilde{L})f)' = \kappa f' \ge 0$.

For (C): $\tilde{L}$ is decreasing, so

$$s\mu f' \le 2s\tilde{L}(1)f\,(s+z-1) = 2sx_0\tilde{L}(1)f \le s(t-1)f.$$

It remains to show that $H \ge s(t-1)$. With $z = 1+x$,

$$H - s(t-1) = x\big[2(1+x)(t-1-x) + s(2t-4-3x)\big].$$

The bracket is affine in s. It is nonnegative at $s = 0$, and at $s = t-1-x$ it equals $(t-1-x)(2t-2-x) \ge 0$. □

4.2. **Regime R3: saturate (C), then solve (D).**

Proposition 4.2. *In R3, the conclusion of Theorem 3.10 holds.*

The construction follows [4, §4.3.3]. Since $h_C = (\log F_C)'$ with

$$F_C := \mu/s^2,$$

we first take $f = F_C$, for which (C) holds with equality. Condition (D) then becomes $d := \widetilde{D}F_C/(2F_C) \le 0$. A direct computation gives $\widetilde{D}F_C = N/s^4$, where in the variable $S = z_0 - z \in (0, x_0)$

$$N = 6\mu_0^2 - 6\mu_0\mu_0' S + c_2S^2 + c_3S^3 + 2(B+4t-15z_0+4)S^5 + 15S^6. \tag{4.1}$$

Here $\mu_0 = \mu(z_0)$, $\mu_0' = \mu'(z_0)$, and c_2, c_3 are affine in (A, B) with explicit polynomial coefficients in (t, z_0). The coefficient of S^4 vanishes.

Lemma 4.3 (One sign change)**.** *In R3, N has exactly one zero $\hat{S} \in (0, x_0)$. Hence $d < 0$ near $z = 1$, $d > 0$ near z_0, and d changes sign exactly once, at $\hat{z} = z_0 - \hat{S}$.*

Proof. Write $B = 3z_0 - A/z_0 + \beta$ with $\beta = L(z_0) > 0$. Substitute $S = x_0 v/(1+v)$, $v \in (0,\infty)$, clear denominators, and let $T_0, \dots, T_6$ be the coefficients of the resulting sextic in v.

- $T_0 = 6z_0^3(z_0-t)^2x_0^2 > 0$ and $T_1 = 6z_0^2(z_0-t)x_0^2(3z_0^2+2z_0-4tz_0-t) > 0$.
- $T_6 = z_0N|_{z=1} = -z_0(t-1)^2x_0^2(p-1) < 0$ in R3, by an exact factorization.

- For $k = 3, 4, 5$, T_k is affine in (A, β) with strictly negative coefficients, and T_{k-1} is affine with nonpositive coefficients. In R3 the admissible set is the quadrant $\{A \geq a_0, \beta \geq 0\}$ with $a_0 = \max(0, z_0((t-1)/(2x_0^2) - 3))$, since the line $p = 1$ passes through its corner. So $\{T_k \geq 0\}$ meets the quadrant in a triangle at the corner, and $T_{k-1} > 0$ on it if two explicit combinations g_1, g_2 of corner values and coefficients are positive. These are rational functions of (t, z_0); we certify $g_1, g_2 > 0$ unconditionally for $k - 1 = 2, 3, 4$ (Appendix A, item C3).

Hence $T_k \leq 0 \Rightarrow T_{k+1} < 0$ for $k = 2, \ldots, 5$, so the sign pattern changes exactly once. By Descartes' rule, N has exactly one root in $(0, x_0)$. □

Proof of Proposition 4.2. Define $f = F_C$ on $(1, \hat{z}]$, and on $[\hat{z}, z_0]$ let f solve $\widetilde{D}f = 0$ with $(f, f')(\hat{z}) = (F_C, F_C')(\hat{z})$. Since $d(\hat{z}) = 0$, f is C^2. Condition (a) holds because $F_C \sim (t-1)(z-1)/x_0^2$ as $z \to 1$. Condition (D) holds on $(1, \hat{z}]$ by Lemma 4.3, and with equality on $[\hat{z}, z_0]$. Condition (C) on $[\hat{z}, z_0]$ follows from the Wronskian argument: with $p_* = \mu \exp(-\int 2sL/\mu)$, the function $W = p_*(F_C f' - F_C' f)$ is non-increasing where $d \geq 0$ and $W(\hat{z}) = 0$.

The seam. It remains to show that $f'(z_0) > 0$. Put $w = \mu f'/f$. It satisfies the Riccati equation

$$\mu w' = R(w) := -w^2 + 2sLw - 2\mu s|L'|, \qquad w(\hat{z}) = \hat{w} := \hat{H}/\hat{s} > 0, \tag{4.2}$$

and w can cross 0 only downward. A function k with $\mu k' \leq R(k)$ in the Dini sense and $k(\hat{z}) \leq \hat{w}$ stays below w. We use $k = \min(\hat{w}, sL)$.

(i) sL is a strict subsolution on all of $(1, z_0)$. Since $(sL)' = -L - s|L'|$, we have $R(sL) - \mu(sL)' = g := s^2L^2 + \mu(L - s|L'|)$. Write $\ell = L(z_0)$ and $a = A/(2z^2z_0)$. Exactly, $L = \ell + s(A/(zz_0) + 3)$ and $L - s|L'| = \ell - As^2/(z^2z_0)$. By (3.4), $L - s|L'| \geq a(z^2 - 2s^2) + 2t$ and $L \geq az(z + 2s) + 2t$. By AM–GM, $(az(z+2s) + 2t)^2 \geq 8t\, az(z+2s)$, so

$$g \;\geq\; 8t\, az(z+2s)s^2 - 2\mu a s^2 + 2t\mu \;\geq\; 2t\mu > 0,$$

because $4t(z + 2s) \geq (z - 1)(t - z)$.

(ii) The constant $\hat{w}$ is a subsolution until sL falls below it. On the interval where $sL \geq \hat{w}$ we have $\hat{w} \leq sL < k_+$, where $k_\pm = sL \pm \sqrt{s^2L^2 - 2\mu s|L'|}$ are the roots of R. So $R(\hat{w}) > 0$ follows from $\hat{w} > k_- = 2\mu s|L'|/k_+$, which in turn follows from

$$\hat{w}L(z) > 2\mu(z)|L'(z)| \qquad (z \in [\hat{z}, z_0)). \tag{4.3}$$

To prove (4.3), note that $Q := \hat{w}L - 2\mu|L'|$ is affine and increasing in B. At the true value $B = B_*(\hat{s})$ (the root condition $N(\hat{s}) = 0$), (3.4) gives $B_* \geq B_m(A) := 3z_0 - A/(2z_0) + 2t$. Since N is decreasing in B, this is the condition $g_N := N(\hat{s}; A, B_m) \geq 0$, and $Q(B_*) = Q(B_m) + \hat{w}\, g_N/|\partial_B N|$. Hence it suffices to prove: for $t > z_0 > 1$, $A \geq 0$, $\hat{s} \in (0, x_0)$ and $z \in [z_0 - \hat{s}, z_0)$,

$$N(\hat{s}; A, B_m(A)) \geq 0 \implies Q(B_m) > 0.$$

Both sides are affine in A, and the coefficient of A in N is negative, so the admissible A form a segment $[0, A_*]$. It therefore suffices to check $A = 0$ and $A = A_*$. After compactifying $(z_0, t, \hat{s}, z)$ to $(0, 1)^4$, these are two polynomial implications of degree at most $(10, 3, 10, 4)$, plus the corresponding statements at $z = \hat{z}$. We certify them by an exact Bernstein covering (Appendix A, item C4).

By (i) and (ii), $k = \min(\hat{w}, sL)$ is a subsolution on $[\hat{z}, z_0)$ with $k(\hat{z}) \leq \hat{w}$. Hence $w > 0$ there. As $z \to z_0$ the defect $g \to \mu(z_0)\ell > 0$, which gives $w(z_0) > 0$, i.e. $f'(z_0) > 0$. So (b) holds, and $f > 0$. □

This completes the proof of Theorem 3.10, and with it Theorem 1.1 for real $\sigma \neq 0$.

5. Zero frequency

Proposition 5.1. *There are no nontrivial outgoing modes at $\sigma = 0$, for $s = \pm 2$.*

Proof. Case $m \neq 0$. Then $w = 0 < w_2$ lies outside the window, and Corollary 3.2 applies. Indeed, $\operatorname{Im}\kappa_1$ and $\operatorname{Im}\kappa_3$ are nonzero because $am \neq 0$.

Case $m = 0$, $s = -2$. The radial equation is $(\Delta_r^{-1}R')' + V\Delta_r^{-2}R = 0$, with $V = -2\Lambda r^2 - 2(1-\alpha) - \lambda$. Put $g = \Delta_r^2 R''''$. The Teukolsky–Starobinsky identity at $K = 0$ shows that g solves the same equation. Moreover g vanishes like Δ_r^2 at both horizons, so the energy identity $\int \Delta_r^{-1}|g'|^2 = \int V\Delta_r^{-2}|g|^2 \leq 0$ gives $g = 0$. Thus R is a cubic polynomial. Inserting a cubic into the equation gives a 4×4 linear system for its coefficients, with determinant $[\lambda^2 + 6(1-\alpha)\lambda + 8 - 4\alpha + 8\alpha^2]^2$. Both roots of the quadratic are negative, and $\lambda \geq \lambda_{\mathrm{cr}} = 2 - 4\alpha > 0$ at $m = \xi = 0$ by Proposition 3.6 and Lemma 2.1. Hence the determinant is positive and $R = 0$.

Case $m = 0$, $s = +2$. The function $\Delta_r^2 R_{+2}$ solves the $s = -2$ equation, so we may repeat the argument.

The two identities used are exact symbolic identities (Appendix A, item C5). □

6. The upper half plane

6.1. Fredholm theory.

Lemma 6.1 (Regular frame)**.** *Let $\chi \in C^\infty(\mathbb{R})$ equal 1 near r_1 and 0 near r_2, and put $b(r) = \chi(r) + (1-\chi(r))/\Delta_r$ on (r_1, r_2). Let (l_K, n_K, m_K) be the Kinnersley tetrad. Then*

$$l = b\,\Delta_r\, l_K, \qquad n = (b\,\Delta_r)^{-1} n_K, \qquad m = m_K$$

extends to a stationary null tetrad that is smooth on $M = [r_1 - \delta, r_2 + \delta] \times S^2$, away from the usual phase ambiguity of m on the axis. In this tetrad, the GHP Teukolsky operator of spin $s = \pm 2$ is a stationary second-order operator with smooth coefficients on sections of a complex line bundle over M, with principal symbol $\frac{1}{2}G(\zeta,\zeta)$, where G is the dual metric. Its outgoing modes are exactly the modes of Lemma 2.2.

Proof. Near r_1, in ingoing coordinates $(v, r, \tilde{\varphi})$,

$$\Delta_r l_K = 2\Xi(r^2 + a^2)\partial_v + 2\Xi a\partial_{\tilde{\varphi}} + \Delta_r\partial_r, \qquad \Delta_r^{-1} n_K = -\partial_r/(2\varrho^2).$$

Both are smooth and nonvanishing. Near r_2, in outgoing coordinates, $l_K = \partial_r + (\text{angular})$ and n_K are already smooth. The factor $b\Delta_r$ is smooth and positive on (r_1, r_2), and equals Δ_r near r_1 and 1 near r_2. The GHP operator is covariant, so its coefficients in this tetrad (spin coefficients and directional derivatives) are smooth. Its principal part is $\text{þ}\text{þ}' - \eth\eth'$, whose symbol is $\zeta(l)\zeta(n) - \zeta(m)\zeta(\bar{m}) = \frac{1}{2}G(\zeta,\zeta)$. The last statement is the definition of outgoing modes, together with Lemma 2.2. □

Fix $m \in \mathbb{Z}$, and let $P_b(\sigma)$ be the spectral family of this operator on $\ker(\partial_{\varphi_*} - im)$, at parameters $b = (\Lambda, M, a)$. The dynamical hypotheses of [6, §5.1] concern only the principal symbol G (radial sets at the horizons, and the global null-bicharacteristic flow). They were verified in the full subextremal range by Petersen–Vasy [11, 10]; see [5, Remark 2.12]. The subprincipal part enters only through the threshold regularity at the radial sets. The trapping hypothesis of [6, §5.1] is needed only for high-energy estimates near the real axis, not for the Fredholm property [6, Theorem 5.4 and the remark in its proof]. We therefore have:

- (F1) $P_b(\sigma)$ is a holomorphic family of Fredholm operators of index 0 on $\operatorname{Im}\sigma > -c$ [6, Theorem 5.4]. Its non-invertibility set is discrete, and it is the set of quasinormal mode (QNM) frequencies.

(F2) There is C_0, uniform for b near b_0, such that $P_b(\sigma)$ is invertible for $\operatorname{Im}\sigma > C_0$. This follows from the global energy estimate on growing function spaces, which uses only hyperbolicity and holds uniformly in b [6, §5.1.2].

(F3) The non-invertibility set is closed in (b,σ), and QNMs move continuously, with constant total rank in small discs [6, Proposition 5.11]. That proof uses only the Fredholm estimates [6, (5.26)] on a precompact set of σ, which hold uniformly in b.

What is *not* available for Teukolsky in the full subextremal range is the high-energy estimate bounding $|\operatorname{Re}\sigma|$. We obtain it sector by sector from the gauge-fixed Einstein operator, for which it is proved in [5].

6.2. **Metric reconstruction.** We use the Geroch–Held–Penrose (GHP) formalism with $\text{þ}' = \Delta - p\gamma - q\bar{\gamma}$ and $\eth' = \bar{\delta} - p\alpha - q\bar{\beta}$ [2]. Let $\mathcal{O}$ be the GHP Teukolsky operator for Ψ_0, and $\mathcal{O}'$ its prime. On any type D Λ-vacuum the Bianchi identities give $\text{þ}\Psi_2 = 3\rho\Psi_2$, $\eth\Psi_2 = 3\tau\Psi_2$ and their primes; there are no Λ-terms. With $P := \Psi_2^{-4/3}$ (GHP type $(0,0)$), and using the formal adjoint rules $\text{þ}^\dagger = -(\text{þ} - \rho - \bar{\rho})$ and $\eth^\dagger = -(\eth - \tau - \bar{\tau}')$, a direct computation gives

$$\mathcal{O}'^\dagger \circ P = P \circ \mathcal{O}. \tag{6.1}$$

Proof of (6.1). Write $\mathcal{O} = (\text{þ} - 4\rho - \bar{\rho})(\text{þ}' - \rho') - (\eth - 4\tau - \bar{\tau}')(\eth' - \tau') + V_0$ and $\mathcal{O}'$ for its prime, where the zeroth-order term V_0 (a constant multiple of Ψ_2, plus a Λ-term) is prime-invariant and real. Using $(AB)^\dagger = B^\dagger A^\dagger$ and the adjoint rules,

$$(\text{þ} - \rho)^\dagger = -(\text{þ} - \bar{\rho}), \qquad (\text{þ}' - 4\rho' - \bar{\rho}')^\dagger = -(\text{þ}' + 3\rho'),$$

and similarly for $\eth$, with $(\rho, \bar{\rho}, \rho')$ replaced by $(\tau, \bar{\tau}', \tau')$. Hence

$$\mathcal{O}'^\dagger = (\text{þ} - \bar{\rho})(\text{þ}' + 3\rho') - (\eth - \bar{\tau}')(\eth' + 3\tau') + V_0.$$

From $\text{þ}\Psi_2 = 3\rho\Psi_2$, $\text{þ}'\Psi_2 = 3\rho'\Psi_2$, $\eth\Psi_2 = 3\tau\Psi_2$ and $\eth'\Psi_2 = 3\tau'\Psi_2$ we get $\text{þ}P = -4\rho P$, $\text{þ}'P = -4\rho' P$, $\eth P = -4\tau P$ and $\eth' P = -4\tau' P$. Therefore

$$(\text{þ} - \bar{\rho})(\text{þ}' + 3\rho')(PX) = (\text{þ} - \bar{\rho})\big(P(\text{þ}' - \rho')X\big) = P(\text{þ} - 4\rho - \bar{\rho})(\text{þ}' - \rho')X,$$

the same computation applies to the angular term, and V_0 commutes with P. This proves (6.1) as an operator identity, with no sampling.

Wald's argument [15], which uses only the decoupling identity $\mathcal{S}'\mathcal{E} = \mathcal{O}'\mathcal{T}'$ and the self-adjointness of the linearized Einstein operator $\mathcal{E}$, then gives the following.

Lemma 6.2 (Reconstruction)**.** *If $\mathcal{O}\Psi = 0$, then $h_\Psi := \mathcal{S}'^\dagger(P\Psi)$ solves $\mathcal{E}h_\Psi = 0$. Moreover, the complex conjugate $\tilde{h} := \overline{h_\Psi}$ satisfies $\Psi_4[\tilde{h}] = c_0(\text{þ}')^4(\bar{P}\bar{\Psi})$ for a universal constant $c_0 \neq 0$ ($c_0 = \frac{1}{2}$ in the normalization of* [8]*).*

Proof. The first claim is Wald's argument combined with (6.1): from $\mathcal{S}'\mathcal{E} = \mathcal{O}'\mathcal{T}'$ and the self-adjointness of $\mathcal{E}$ we get $\mathcal{E}\mathcal{S}'^\dagger = \mathcal{T}'^\dagger\mathcal{O}'^\dagger$, and $\mathcal{O}'^\dagger(P\Psi) = P\mathcal{O}\Psi = 0$.

For the second claim we use the convention $h_{xy} := h_{ab}x^a y^b$. The operator $\mathcal{S}'$ only involves the components T_{nn}, $T_{n\bar{m}}$ and $T_{\bar{m}\bar{m}}$. So $\mathcal{S}'^\dagger$ produces a tensor of the form $h_\Psi = X\, n\otimes n + Y\, n\odot\bar{m} + Z\,\bar{m}\otimes\bar{m}$ (indices lowered), whose only nonzero components are h_{ll}, h_{lm} and h_{mm}. By GHP type counting, the linearized Ψ_4 depends only on h_{nn}, $h_{n\bar{m}}$ and $h_{\bar{m}\bar{m}}$. Hence $\Psi_4[h_\Psi] = 0$. For $\tilde{h} = \overline{h_\Psi}$ the only nonzero components are h_{ll}, $h_{l\bar{m}}$ and $h_{\bar{m}\bar{m}}$, so $\Psi_4[\tilde{h}]$ comes from $\tilde{h}_{\bar{m}\bar{m}} = \overline{h_{mm}}$ alone. The resulting expression involves only $\text{þ}'$-derivatives and the coefficients $\rho', \bar{\rho}'$. Its evaluation uses only the Ricci identities $\text{þ}'\rho' = \rho'^2$ and $\text{þ}'\bar{\rho}' = \bar{\rho}'^2$, so no commutator and no Λ-term enters. The algebra is therefore literally that of the Kerr case [15, 8], which gives $\Psi_4[\tilde{h}] = c_0(\text{þ}')^4(\bar{P}\bar{\Psi})$. Only $c_0 \neq 0$ is used below. □

6.3. **The kernel lemma.** Let $\kappa_2^\sharp = |\Delta_r'(r_2)|/(2\Xi\rho_2)$ be the normalized surface gravity of the cosmological horizon, and define the *cosmological ladder*

$$\mathcal{Z}_m = \{m\Omega_2 + iN\kappa_2^\sharp : \ N \in \{1,2,3,\dots\}\}.$$

Lemma 6.3 (Kernel lemma)**.** *Let* $\operatorname{Im}\sigma > 0$ *and* $\sigma \notin \mathcal{Z}_m$*. If* $F = e^{-i\sigma t_* + im\varphi_*}v$ *is a mode of GHP type* $(0,4)$ *or* $(4,0)$*, regular at the future cosmological horizon, with* $(\text{þ}')^4 F = 0$*, then* $F = 0$*.*

Proof. In the tetrad of Lemma 6.1, þ′ differentiates only along n, which has no ∂_θ-component. So for each fixed θ, $(\text{þ}')^4 F = 0$ is a fourth-order linear ODE in r, the composition of four first-order operators

$$n(\cdot) - (p_i + q_i)\gamma, \qquad p_i + q_i \in \{4,2,0,-2\}.$$

On modes, $n = -(\Delta_r/2\varrho^2)\big(\partial_r + iK/\Delta_r\big)$ in the Kinnersley normalization, and $\gamma = \mu + \Delta_r'/(4\varrho^2)$ with μ vanishing like Δ_r. Hence each first-order factor has kernel spanned by $\exp(-i\int K/\Delta_r)$ times Δ_r^k times an analytic function, with $k \in \frac{1}{2}(p_i+q_i) + \mathbb{Z} \subset \mathbb{Z}$. Every local solution near r_2 is therefore a finite sum of terms $(r_2 - r)^{-\kappa_2 + k}\log^j(r_2 - r)$ times analytic functions, with $k \in \mathbb{Z}$.

Regularity at the future cosmological horizon is the branch $(r_2 - r)^{\kappa_2} C^\infty$ (Lemma 2.2). So a nonzero F requires $2\kappa_2 \in \mathbb{Z}$. For $\operatorname{Im}\sigma > 0$ we have $\operatorname{Re}(2\kappa_2) = 2\Xi\rho_2 \operatorname{Im}\sigma/|\Delta_r'(r_2)| > 0$ and $\operatorname{Im}(2\kappa_2) = 2\Xi(\rho_2 \operatorname{Re}\sigma - am)/\Delta_r'(r_2)$. Hence $\operatorname{Re}\sigma = m\Omega_2$ and $\operatorname{Im}\sigma = N\kappa_2^\sharp$ with $N = 2\kappa_2 \in \{1,2,3,\dots\}$. Finally, F vanishes identically once it vanishes near r_2, by uniqueness for the ODE. □

Proposition 6.4 (Transfer)**.** *Let* Ψ *be an outgoing* $s = +2$ *Teukolsky mode at frequency* σ *in the sector* m*, with* $\operatorname{Im}\sigma > 0$ *and* $\sigma \notin \mathcal{Z}_m$*. Then* $\tilde{h}$ *from Lemma 6.2 is a smooth outgoing mode of the linearized Einstein equation, at frequency* $-\bar{\sigma}$ *in the sector* $-m$*, with* $\Psi_4[\tilde{h}] \neq 0$*. The case* $s = -2$ *is the same with the roles of* l *and* n*, and of the two horizons, exchanged: one uses the ingoing gauge and the ladder* $\mathcal{Z}_m^e$ *of the event horizon.*

Proof. Smoothness. $\mathcal{S}'^\dagger$ is GHP-covariant. Near each future horizon it may therefore be evaluated in a principal tetrad that is smooth there. In such a tetrad, outgoing modes are smooth, so $\tilde{h}$ is smooth.

Nonvanishing of Ψ_4. The function $\bar{P}\bar{\Psi}$ is regular at both horizons, and its frequency $-\bar{\sigma}$ lies off $\mathcal{Z}_{-m} = -\overline{\mathcal{Z}_m}$. By Lemma 6.3, $(\text{þ}')^4(\bar{P}\bar{\Psi}) \neq 0$. □

6.4. **Compactness, sector by sector.**

Lemma 6.5. *Let* b_0 *be subextremal and* $m \in \mathbb{Z}$*. There are a neighborhood* $U \ni b_0$ *and* $R_m < \infty$ *such that, for all* $b \in U$*, every Teukolsky QNM of spin* ± 2 *in the sector* m *with* $\operatorname{Im}\sigma \geq 0$ *satisfies* $|\sigma| \leq R_m$*.*

Proof. Let L_b be the linearized gauge-fixed Einstein operator of [5, (23)], with the constraint damping fixed at b_0. By [5, Remark 2.12 and §2.5], the operators L_b, $b \in U$, form a continuous family of stationary, principally scalar operators satisfying the assumptions of [6, §5.1], including the trapping condition (by [5, Theorem 3.1]). By [6, §5.1.2], these assumptions then hold uniformly near b_0, and the high-energy estimates [6, (5.7), (5.8)] hold with constants uniform in $b \in U$. In particular there is R_0 such that every QNM of L_b with $\operatorname{Im} \geq 0$ has $|\sigma_T| \leq R_0$ for all $b \in U$. Here σ_T is the T-frequency, which differs from the ∂_t-frequency by $m'\Omega_T$ on the sector m'.

Let Ψ be a QNM in the sector m with $\operatorname{Im}\sigma \geq 0$. If $\operatorname{Im}\sigma = 0$ there is nothing to prove, by Theorem 1.1 on the real axis. By (F2), $\operatorname{Im}\sigma \leq C_0$, so the ladder points in the strip $0 < \operatorname{Im}\sigma \leq C_0$ satisfy $|\sigma| \leq |m|\Omega_2 + C_0$, or $|\sigma| \leq |m|\Omega_1 + C_0$ for the event-horizon ladder. If σ is not a ladder point, Proposition 6.4 gives $\tilde{h}$ at frequency $-\bar{\sigma}$ with $\Psi_4[\tilde{h}] \neq 0$.

Next we gauge-fix $\tilde{h}$: put $h' = \tilde{h} + \delta^*\omega$, where ω solves the linearized gauge condition $D\Upsilon(\delta^*\omega) = -D\Upsilon(\tilde{h})$ at frequency $-\bar{\sigma}$. The operator $\omega \mapsto D\Upsilon(\delta^*\omega)$ is a principally scalar wave operator on 1-forms, so it is Fredholm of index 0 by (F1) in its version for 1-forms.

- If it is invertible at $-\bar{\sigma}$, then h' is a QNM of L_b, and $h' \neq 0$ since $\Psi_4[h'] = \Psi_4[\tilde{h}] \neq 0$.
- If it is not invertible, there is $\omega \neq 0$ with $D\Upsilon(\delta^*\omega) = 0$. Then $L_b(\delta^*\omega) = 0$, and $\delta^*\omega \neq 0$ because KdS has no Killing modes with Im $\neq 0$. So $\delta^*\omega$ is a QNM of L_b at frequency $-\bar{\sigma}$.

In either case L_b has a QNM at T-frequency $-\bar{\sigma} - (-m)\Omega_T$, so $|\sigma| \leq R_0 + |m|\Omega_T$. For $s = -2$ the event-horizon ladder enters instead. Hence we can take $R_m = \max(R_0 + |m|\Omega_T,\ |m|\Omega_1 + C_0,\ |m|\Omega_2 + C_0)$. □

6.5. **The base case.**

Lemma 6.6 (Schwarzschild–de Sitter)**.** *For all but countably many values of* $\Lambda M^2 \in (0, 1/9)$*, SdS has no Teukolsky QNMs of spin* ± 2 *with* $\operatorname{Im}\sigma > 0$.

Proof. Let $a = 0$. The angular operator then does not depend on σ or Λ, so modes separate into spin-weighted spherical harmonics for every $\sigma \in \mathbb{C}$, with a discrete real spectrum $\{\lambda_\ell = \ell(\ell+1) - 4\}_{\ell \geq 2}$ (in the convention of Proposition 3.6) independent of Λ. Two exact computations hold on separated modes, in the Kinnersley tetrad with $D^\dagger = \partial_r + i\sigma r^2/\Delta_r$ and $D = \partial_r - i\sigma r^2/\Delta_r$ (Appendix A, item C8):

$$(\text{þ}')^4(r^4 R) = \frac{1}{16 r^4}\,\Delta_r^2 (D^\dagger)^4 \Delta_r^2 R, \qquad D^4 \Delta_r^2 (D^\dagger)^4 \Delta_r^2 R = \mathcal{C}_\ell(\sigma) R, \tag{6.2}$$

where $\mathcal{C}_\ell(\sigma) = [(\lambda_\ell + 2)(\lambda_\ell + 4)]^2 + 144 M^2 \sigma^2$, the second identity holds for every solution R of the $s = +2$ radial equation, and Λ drops out. Since $P \propto r^4$ at $a = 0$, the first identity shows that $(\text{þ}')^4(P\Psi)$ vanishes only if $\Delta_r^2 (D^\dagger)^4 \Delta_r^2 R = 0$, hence only if $\mathcal{C}_\ell(\sigma) = 0$, that is, $\sigma = \sigma_\ell := i(\lambda_\ell + 2)(\lambda_\ell + 4)/(12M)$.

By Lemma 6.3 (with $\Omega_2 = 0$), this can happen for a mode regular at both horizons only if $\sigma_\ell = iN\kappa_2^\sharp$ for some integer $N \geq 1$, that is, only if $M\kappa_2^\sharp \in \{(\lambda_\ell + 2)(\lambda_\ell + 4)/(12N)\}$. The right-hand side is a countable set independent of Λ. On the other hand, $M\kappa_2^\sharp$ is a nonconstant real-analytic function of $\Lambda M^2 \in (0, 1/9)$, so this happens only for countably many values of ΛM^2.

Off this set, Lemma 6.2 and the argument of Proposition 6.4 produce, from a QNM with $\operatorname{Im}\sigma > 0$, a linearized Einstein mode with Im > 0 and $\Psi_4 \neq 0$. By [6, §7], every such mode on SdS is pure gauge, and pure gauge modes have $\Psi_4 = 0$. This is a contradiction. The case $s = -2$ is the same, using the ingoing gauge, þ^4, and the event horizon. □

6.6. **Conclusion.**

Proof of Theorem 1.1 for $\operatorname{Im}\sigma > 0$. Fix a subextremal b_1 and $m \in \mathbb{Z}$. Choose an SdS point b_* as in Lemma 6.6, and a continuous path $p : [0, 1] \to \{\text{subextremal parameters}\}$ with $p(0) = b_*$ and $p(1) = b_1$. For example, scale r_0 to 0 with r_1, r_2 fixed, as in [4, §6], and then move within the SdS family. Let S be the set of $s \in [0, 1]$ for which the sector m has a QNM with $\operatorname{Im}\sigma > 0$.

- S is open, by (F3).
- S is closed. Let $s_j \to s_0$ with $s_j \in S$ and QNMs σ_j. By Lemma 6.5 and (F2), the σ_j lie in the compact set $\{0 \leq \operatorname{Im}\sigma \leq C_0,\ |\sigma| \leq R_m\}$. A subsequence converges to some σ_0, which is a QNM at $p(s_0)$ by (F3). Now $\sigma_0 \neq 0$, because $P(0)$ is invertible by Proposition 5.1 and invertibility is open. And $\sigma_0 \notin \mathbb{R}$ by §4 and Corollary 3.2. Hence $\operatorname{Im}\sigma_0 > 0$.
- $0 \notin S$, by Lemma 6.6.

Here (F2), (F3) and Lemma 6.5 are applied in finitely many neighborhoods U covering the compact path. Hence $S = \emptyset$. Generalized modes reduce to modes, since their top coefficient is a mode. □

7. Ungauged linearized Einstein mode stability

Throughout this section $a \neq 0$, and $\Omega_T = a/(r_2^2 + a^2)$, so that $T = \partial_t + \Omega_T \partial_\varphi$ in Boyer–Lindquist coordinates. Let h be as in Theorem 1.2. Its Fourier components in φ_* are generalized $(\partial_t, \partial_\varphi)$-modes of type $(\sigma + m\Omega_T, m)$, whose imaginary part is $\operatorname{Im}\sigma \geq 0$. The linearized Teukolsky scalars $\delta\Psi_0$ and $\delta\Psi_4$ are gauge and tetrad invariant on type D, and they are outgoing generalized Teukolsky modes. By Theorem 1.1, applied to each Fourier component of h and of $\bar{h}$, the scalars $\delta\Psi_0$ and $\delta\Psi_4$ vanish on the exterior $r_1 < r < r_2$. The next lemma extends this to all of M:

$$\delta\Psi_0 = \delta\Psi_4 = 0 \quad \text{on } M.$$

Lemma 7.1 (Collars)**.** *Let w be a smooth generalized mode on M of a stationary, principally scalar operator with smooth coefficients and principal symbol $\frac{1}{2}G$, for instance the Teukolsky operator of Lemma 6.1. If $w = 0$ on $r_1 < r < r_2$, then $w = 0$ on M.*

Proof. Consider the collar $r_1 - \delta < r < r_1$. There dr is timelike, and r decreases towards the future. Let p lie in the collar. Every past-directed causal curve from p has strictly increasing r. The set $D_p = J^-(p) \cap \{r \leq r_1\}$ is compact, since in ingoing coordinates the past cone of p meets the event horizon in a compact set. Its boundary consists of a compact piece of the event horizon and a piece of the past light cone of p. Since w is smooth and vanishes for $r > r_1$, it vanishes to infinite order on the horizon. The standard energy estimate on D_p involves fluxes through a null boundary that are tangential and hence vanish, and fluxes through the cone of p that are nonnegative [3, §23.2]. It gives $w = 0$ near p. The collar $r_2 < r < r_2 + \delta$ is treated in the same way, with time reversed relative to r and the cosmological horizon in place of the event horizon. □

7.1. Linearized Goldberg–Sachs.

Lemma 7.2. *After an infinitesimal null rotation of the tetrad, $\delta\Psi_1 = \delta\Psi_3 = 0$, and then $\delta\kappa = \delta\sigma = \delta\kappa' = \delta\sigma' = 0$.*

Proof. To first order, a null rotation about n with parameter c shifts $\delta\Psi_1$ by $3c\Psi_2$ and fixes $\delta\Psi_0$ and $\delta\Psi_2$; the analogous statement holds about l. Since $\Psi_2 \neq 0$, the choice is algebraic. The Λ-vacuum Bianchi identities contain no Λ-terms. Linearizing

$$\text{þ}\Psi_1 - \eth'\Psi_0 = -\tau'\Psi_0 + 4\rho\Psi_1 - 3\kappa\Psi_2, \qquad \text{þ}'\Psi_0 - \eth\Psi_1 = \rho'\Psi_0 - 4\tau\Psi_1 + 3\sigma\Psi_2$$

gives $\delta\kappa = \delta\sigma = 0$; priming gives the rest. □

All perturbed quantities below are local linear functions of h. They are boost-spin invariant and commute with $\mathcal{L}_{\partial_t}$ and $\mathcal{L}_{\partial_\varphi}$.

7.2. **The Killing-spinor field.** On a type D Λ-vacuum with $\kappa = \sigma = \kappa' = \sigma' = 0$, the following identities hold, together with their primes:

$$\text{þ}\rho = \rho^2, \quad \eth\rho = (\rho - \bar{\rho})\tau, \quad \text{þ}\tau = (\tau - \bar{\tau}')\rho, \quad \eth\tau = \tau^2,$$
$$\text{þ}\rho' - \text{þ}'\rho = \tau\bar{\tau} - \tau'\bar{\tau}', \quad \eth\tau' - \eth'\tau = \rho\bar{\rho}' - \bar{\rho}\rho', \quad \text{þ}\tau' = \eth'\rho, \quad \text{þ}'\tau = \eth\rho'.$$

They follow from the GHP Ricci identities and commutators. We verified them exactly on KdS, and on the accelerating NUT Plebański–Demiański metric, which has $\tau \neq -\bar{\tau}'$ (Appendix A, item C6).

Let $\phi = \Psi_2^{-1/3}$ (normalized so that $\phi = r - ia\cos\theta$ on KdS), and put

$$A = \phi\rho, \quad B = -\phi\rho', \quad C = \phi\tau, \quad D = -\phi\tau', \qquad \xi = Bl + An - Dm - C\bar{m}.$$

Lemma 7.3. *The ten Killing equations for ξ are algebraic consequences of the identities above. On KdS, $\xi = -\Xi\,\partial_t$.*

Linearizing Lemma 7.3 gives $\mathcal{L}_\xi h + \mathcal{L}_{\delta\xi} g = 0$, where $\delta\xi = \delta\xi[h]$.

Lemma 7.4 (Reality)**.** *Put $J[h] = \big(\delta\xi[h] - \overline{\delta\xi[\bar{h}]}\big)/(2i)$. Then $\mathcal{L}_{J[h]} g = 0$. If $(\sigma, m) \neq (0,0)$, then $J[h] = 0$.*

Proof. For real h, $J[h] = \operatorname{Im}\delta\xi$, and the imaginary part of the linearized Killing equation reads $\mathcal{L}_J g = 0$; by complex linearity this holds for all h.

The local Killing algebra of KdS with $a \neq 0$ is $\operatorname{span}\{\partial_t, \partial_\varphi\}$ on every open set. Indeed, a Killing field preserves Ψ_2, hence r and $a\cos\theta$. It is therefore tangent to the (t,φ)-orbits, and the isotropy is trivial because it must fix ∇r, $\nabla(a\cos\theta)$ and the principal null directions.

Moreover, J is complex linear and commutes with $\mathcal{L}_{\partial_t}$ and $\mathcal{L}_{\partial_\varphi}$. For a mode of type $(\sigma, m) \neq (0,0)$, it is therefore a constant combination of $\partial_t, \partial_\varphi$ of that type, hence zero. □

Remark 7.5. The same argument shows $J[h] = 0$ in two further cases. If $(\mathcal{L}_T + i\sigma)h = 0$ with $\sigma \neq 0$, then $\mathcal{L}_T J[h] = -i\sigma J[h]$, while $\mathcal{L}_T$ annihilates every Killing field of KdS. If $\mathcal{L}_T h = 0$ and h has vanishing φ-average, then $J[h]$ is a constant combination of $\partial_t, \partial_\varphi$, hence equal to its own φ-average, which is J of the φ-average of h, namely 0.

Consequently $\phi\rho = \bar{\phi}\bar{\rho}$, $\phi\rho' = \bar{\phi}\bar{\rho}'$ and $\phi\tau = -\bar{\phi}\bar{\tau}'$ hold to first order.

7.3. **Carter's Killing field.** Let $R = \operatorname{Re}\phi$ and $I = \operatorname{Im}\phi$. These are the Carter coordinates r and $-a\cos\theta$. Reality gives

$$\text{Þ}R = -A, \quad \text{Þ}'R = B, \quad \text{ð}R = 0, \quad \text{Þ}I = \text{Þ}'I = 0, \quad \text{ð}I = iC.$$

Define

$$\zeta = I^2(Bl + An) + R^2(Dm + C\bar{m}).$$

Lemma 7.6. *To first order, ζ is Killing. On KdS, $\zeta = -\Xi(a^2\partial_t + a\partial_\varphi)$.*

Proof. The ll, nn, mm and $\bar{m}\bar{m}$ components are immediate from those of ξ.

- The ln component equals $-|\phi|^2\phi(\tau\bar{\tau} - \tau'\bar{\tau}') = 0$, since by reality $\tau\bar{\tau} = \tau'\bar{\tau}'$.
- The $m\bar{m}$ component equals $|\phi|^2\phi(\rho\bar{\rho}' - \rho'\bar{\rho}) = 0$, again by reality.
- The lm component equals $2AC\phi + |\phi|^2(\text{ð}A + \bar{\tau}'A) = 0$, using $\text{ð}A = -\phi\bar{\rho}\tau$.
- The nm component equals $-2BC\bar{\phi} + |\phi|^2(\text{ð}B + \tau B)$. Differentiating the first-order reality relation $\phi\rho' = \bar{\phi}\bar{\rho}'$ along m gives $\text{ð}B + \tau B = 2\tau B$, and the sum vanishes.

The remaining components are complex conjugates. The value on KdS is an exact computation (item C6). □

7.4. **Pure gauge modes.** On KdS, $\partial_t = -\xi/\Xi$ and $\partial_\varphi = (a^2\xi - \zeta)/(\Xi a)$. Hence there are constants α_T, β_T and $\alpha_\varphi, \beta_\varphi$ with

$$T = \alpha_T\xi + \beta_T\zeta, \qquad \partial_\varphi = \alpha_\varphi\xi + \beta_\varphi\zeta.$$

Whenever $J[h] = 0$, Lemmas 7.3 and 7.6 give $\mathcal{L}_\xi h = -\mathcal{L}_{\delta\xi} g$ and $\mathcal{L}_\zeta h = -\mathcal{L}_{\delta\zeta} g$, and therefore

$$\mathcal{L}_T h = -\mathcal{L}_{Y_T} g, \qquad \mathcal{L}_{\partial_\varphi} h = -\mathcal{L}_{Y_\varphi} g, \qquad Y_\bullet = \alpha_\bullet\delta\xi + \beta_\bullet\delta\zeta. \tag{7.1}$$

The vector fields Y_T and Y_φ are local linear expressions in h. They are smooth across the horizons: ξ, ζ, $\delta\xi$ and $\delta\zeta$ are boost-spin invariant constructions from the perturbed Weyl spinor, so they can be computed in a horizon-regular principal tetrad, and $\Psi_2 \neq 0$ on $r > 0$.

Proposition 7.7. *(i) If $(\mathcal{L}_T + i\sigma)h = 0$ with $\sigma \neq 0$, then $h = \mathcal{L}_X g$ with $X = Y_T/(i\sigma)$.*
(ii) If $\mathcal{L}_T h = 0$ and h has vanishing φ-average, then $h = \mathcal{L}_X g$, where X is the unique φ-antiderivative of $-Y_\varphi$ with vanishing φ-average.
In both cases X is smooth, and $(\mathcal{L}_T + i\sigma)X = 0$.

Proof. In both cases $J[h] = 0$ by Remark 7.5, so (7.1) holds.
(i) Here $-i\sigma h = \mathcal{L}_T h = -\mathcal{L}_{Y_T} g$.
(ii) The field Y_φ has vanishing φ-average, because Y_φ commutes with rotations and h has vanishing average. So its mean-zero φ-antiderivative X (with $[\partial_\varphi, X] = -Y_\varphi$) is smooth. Then $\mathcal{L}_{\partial_\varphi}(h - \mathcal{L}_X g) = -\mathcal{L}_{Y_\varphi} g + \mathcal{L}_{Y_\varphi} g = 0$. So $h - \mathcal{L}_X g$ is axisymmetric with vanishing average, hence zero. □

7.5. **The stationary axisymmetric sector.**

Proposition 7.8. *If $\mathcal{L}_{\partial_t} h = \mathcal{L}_{\partial_\varphi} h = 0$ and $\delta\Psi_0 = \delta\Psi_4 = 0$, then $h = c_M \partial_M g + c_a \partial_a g + \mathcal{L}_X g$, with X affine in (t, φ).*

Proof. Step 1: restoring reality. Here $J[h] = A'\partial_t + B'\partial_\varphi$ need not vanish. A constant phase change of ϕ removes A'. Subtracting a multiple of the tangent to the Plebański–Demiański family in the direction of the acceleration removes B': this is the only Plebański–Demiański tangent that makes ξ complex, and its J has nonzero ∂_φ-part (exact nonlinear check, item C6). After this, reality holds, and ξ and ζ are first-order Killing fields. They commute, since $\delta\xi$ and $\delta\zeta$ are background Killing fields.

Step 2: Carter's ansatz. One has $\nabla R \perp \nabla I$. Put $\Delta_R := -2AB|\phi|^2$ and $\Delta_I := -2CD|\phi|^2$. Then Δ_R depends on R alone and Δ_I on I alone, since $\eth(AB|\phi|^2) = 0$ by reality and $\xi(\Delta_R) = 0$. All the inner products of $\xi, \zeta, \nabla R$ and ∇I are functions of $(R, I, \Delta_R, \Delta_I)$. In coordinates adapted to (ξ, ζ, R, I), the perturbed metric is therefore, to first order, Carter's separable ansatz

$$\frac{S}{\Delta_R} dR^2 + \frac{S}{\Delta_I} dI^2 + \frac{\Delta_I (d\tilde{t} + R^2 d\psi)^2 - \Delta_R (d\tilde{t} - I^2 d\psi)^2}{S}, \qquad S = R^2 + I^2.$$

Step 3: the field equations. On this ansatz the Λ-vacuum equations are linear in (Δ_R, Δ_I), with general solution

$$\Delta_R = -\tfrac{\Lambda}{3} R^4 + \varepsilon R^2 - 2\mathfrak{m} R + \gamma, \qquad \Delta_I = -\tfrac{\Lambda}{3} I^4 - \varepsilon I^2 + 2\mathfrak{n} I + \gamma$$

(exact computation, item C7). Modulo the rescaling gauge, the variations are δM, δa and the NUT variation $\delta\mathfrak{n}$, together with the acceleration removed in Step 1.

Step 4: axis regularity. Write the background as $\Delta_I = (a^2 - I^2)(c + \tilde{\lambda} I^2)$ with $\tilde{\lambda} = \Lambda/3$ and $c > 0$, so the axes are the roots $I = \pm a$ and $\Delta_I'(\pm a) = \mp 2a(c + \tilde{\lambda} a^2)$. Two conditions are needed.

(i) Misner string. At a simple root I_0 of Δ_I, the Killing field that vanishes on the axis is $\partial_\psi + I_0^2 \partial_{\tilde{t}}$, as one reads off from the ansatz. Under $\delta\Delta_I = 2\,\delta\mathfrak{n}\, I$ both roots move by the same amount: $\delta I_\pm = -2\delta\mathfrak{n}\, I_\pm / \Delta_I'(I_\pm) = \delta\mathfrak{n}/(c + \tilde{\lambda} a^2)$, since Δ_I' is odd. So the two axis generators differ by $4a\,\delta I_+\,\partial_{\tilde{t}}$. Because $\tilde{t}$ is not periodic on M, both axes are regular only if the generators coincide. For $a \neq 0$ this forces $\delta\mathfrak{n} = 0$. For $a = 0$ this is the classical Misner-string condition on Schwarzschild–NUT–de Sitter, with the same conclusion. The acceleration tangent does not move the axis generators at first order, since its $\delta\Delta_I$ is even in I after the conformal rescaling.

(ii) Conical regularity. With $\delta\mathfrak{n} = 0$, the ratio of the two conical periods along the acceleration tangent is $1 + 4M\,\delta\alpha_{\mathrm{acc}} + O(\delta^2)$. This is the first-order expansion of the C-metric condition $P(0) = P(\pi)$ in Griffiths–Podolský form, $P = 1 - 2\alpha M \cos\theta + O(a^2, \Lambda a^2)\cos^2\theta$, where the $\cos^2\theta$ term is even. A single period of φ regularizes both axes only if $\delta\alpha_{\mathrm{acc}} = 0$.

The system for $(\delta\mathfrak{n}, \delta\alpha_{\mathrm{acc}})$ is triangular, with diagonal entries $4a/(c + \tilde{\lambda} a^2)$ and $4M$, both nonzero, so both variations vanish. The gauge vector X generates the change from the background to the perturbed adapted coordinates, together with the rescalings of ξ and ζ.

Step 5: the gauge vector on all of M. Here $\partial_M g$ and $\partial_a g$ denote the parameter derivatives of the KdS family in the regular coordinates $(t_*, r, \theta, \varphi_*)$, as in [5].

Steps 1–4, applied on each of the three regions of $M \setminus \{r = r_1, r_2\}$, give constants $(c_M^{(i)}, c_a^{(i)})$ and smooth $X^{(i)}$ with $h = c_M^{(i)}\partial_M g + c_a^{(i)}\partial_a g + \mathcal{L}_{X^{(i)}} g$ there. The constants agree. Indeed, for $K \in \{\partial_t, \zeta\}$ the Iyer–Wald form $\delta Q_K - K \cdot \Theta$ evaluated on h is closed and smooth on M, so its integral $\delta H_K[h]$ over a sphere $\{t_* = \tau, r = \rho\}$ is the same for all ρ, including spheres in different regions. On $\mathcal{L}_X g$ it vanishes (see the proof of Theorem 1.2). So $\delta H_K[c_M^{(i)}\partial_M g + c_a^{(i)}\partial_a g]$ is independent of i, and since (E, J) has nonzero Jacobian with respect to (M, a), so is $(c_M^{(i)}, c_a^{(i)}) =: (c_M, c_a)$. (An earlier version used $\delta(\Psi_2\phi^3)$, which vanishes identically because ϕ is normalized by Ψ_2.)

By Lemma 7.1, Steps 1–4 apply verbatim on each of the three regions, where $\Psi_2 \neq 0$ and $\Delta_R \neq 0$. So $k := h - c_M\partial_M g - c_a\partial_a g$ is smooth on M, and $k = \mathcal{L}_{X^{(i)}} g$ on the i-th region with $X^{(i)}$ smooth there.

We now glue these. Write $X = X^r\partial_r + X^\theta\partial_\theta + Y$ with $Y = Y^t\partial_{t_*} + Y^\varphi\partial_{\varphi_*}$.

- *The (r, θ)-components.* Since $R = \operatorname{Re}\phi$ and $I = \operatorname{Im}\phi$ are natural functions, $X(R) = \delta R[k]$ and $X(I) = \delta I[k]$. These are local linear expressions in k, smooth on M. Hence X^r and X^θ are given by one formula on all three regions, and they are smooth on M.
- *The remaining components.* Let $k_1 = k - \mathcal{L}_{X^r\partial_r + X^\theta\partial_\theta} g$, which is smooth on M. On each region, $Y^\bullet = c^\bullet t_* + f^\bullet(r, \theta)$ with constants $c^\bullet$. Since $\partial_{t_*}, \partial_{\varphi_*}$ are Killing,

$$k_1 = \mathcal{L}_Y g = 2\, dY^t \odot K_t^\flat + 2\, dY^\varphi \odot K_\varphi^\flat, \qquad K_t = \partial_{t_*},\ K_\varphi = \partial_{\varphi_*}.$$

  Claim: at every point off the axis, the linear map $(dY^t, dY^\varphi) \mapsto \mathcal{L}_Y g$, restricted to pairs of covectors annihilating ∂_{φ_*}, is injective. Indeed, $K_t^\flat, K_\varphi^\flat$ are linearly independent, and $\alpha \odot K_t^\flat + \beta \odot K_\varphi^\flat = 0$ forces $\alpha = yK_\varphi^\flat$ and $\beta = -yK_t^\flat$. Then $\alpha(\partial_{\varphi_*}) = y\, g_{\varphi\varphi} = 0$ gives $y = 0$.

  Hence, off the axis, (dY^t, dY^φ), and with it $(c^\bullet, df^\bullet)$, is given by one smooth formula in k_1, the same on all three regions. Integrating, Y is smooth on M minus the axis, up to additive constants, which are Killing. Near the axis, Y is smooth on each region. It extends smoothly across the codimension-three set where the axis meets the horizons by Killing transport: $(Y, \nabla Y)$ satisfies a linear ODE with coefficients smooth in (g, k_1) along any curve.

So X is smooth on M. Then $k - \mathcal{L}_X g$ is smooth on M and vanishes on each of the three regions, hence on M. The same argument applies to the t_*-linear parts that arise in the generalized-mode induction below. □

7.6. **Generalized modes and proof of Theorem 1.2.**

Proof of Theorem 1.2. Part (1). Let $(\mathcal{L}_T + i\sigma)^k h = 0$ with $\sigma \neq 0$. We argue by induction on k; the case $k = 1$ is Proposition 7.7(i). For $k \geq 2$, the top coefficient $h_1 = (\mathcal{L}_T + i\sigma)^{k-1} h$ is a mode, so $h_1 = \mathcal{L}_{X_1} g$ with $(\mathcal{L}_T + i\sigma)X_1 = 0$. Put $Y = t_*^{k-1} X_1/(k-1)!$, which is smooth since t_* is regular. Then $(\mathcal{L}_T + i\sigma)^{k-1} Y = X_1$, so $h - \mathcal{L}_Y g$ has order $k - 1$.

Part (2). Let $(\mathcal{L}_T)^k h = 0$, and split $h = h_0 + h_\perp$ into its φ-average and the rest. The same induction, using Proposition 7.7(ii), shows that $h_\perp$ is pure gauge. The average h_0 satisfies $\mathcal{L}_{\partial_\varphi} h_0 = 0$ and $(\mathcal{L}_{\partial_t})^k h_0 = 0$. For $k = 1$, Proposition 7.8 applies. For $k \geq 2$:

- Let $h_1 := (\mathcal{L}_{\partial_t})^{k-1} h_0$. It is stationary and axisymmetric and solves the linearized equation, so by Proposition 7.8 $h_1 = c_M\partial_M g + c_a\partial_a g + \mathcal{L}_X g$. By Step 5 of that proof, $X = X_0 + t_* K$, where X_0 is stationary and $K = c^t\partial_{t_*} + c^\varphi\partial_{\varphi_*}$ is Killing.
- The Iyer–Wald charges δH_K, for $K \in \{\partial_t, \zeta\}$, are independent of the 2-sphere. They vanish on $\mathcal{L}_Y g$ for every smooth Y: from $\Theta(\mathcal{L}_Y g) = Y \cdot L + dQ_Y$ and naturality of Q,

one gets $\delta H_K[\mathcal{L}_Y g] = -\oint(Q_{[Y,K]} + Q_{[K,Y]}) = 0$. Since $h_1 = (\mathcal{L}_{\partial_t})^{k-1}h_0$ with $k \geq 2$, we have $\delta H_K[h_1] = \frac{d}{d\tau}\delta H_K[(\mathcal{L}_{\partial_t})^{k-2}h_0]|_{\{t_*=\tau\}} = 0$, because the charge of a solution does not depend on τ. Hence $\delta H_K[c_M\partial_M g + c_a\partial_a g] = 0$. Since $(E, J) = (M/\Xi^2, aM/\Xi^2)$ has Jacobian $M/\Xi^4 \neq 0$ with respect to (M, a), it follows that $c_M = c_a = 0$.

- Put $Z = \frac{t_*^{k-1}}{(k-1)!}X_0 + \frac{t_*^k}{k!}K$, which is smooth since t_* is regular. Because $[\partial_{t_*}, X_0] = [\partial_{t_*}, K] = 0$ and $\partial_{t_*}t_* = 1$, we get $(\mathrm{ad}_{\partial_{t_*}})^{k-1}Z = X_0 + t_*K = X$. Hence $(\mathcal{L}_{\partial_t})^{k-1}(h_0 - \mathcal{L}_Z g) = h_1 - \mathcal{L}_X g = 0$, so $h_0 - \mathcal{L}_Z g$ has order $k-1$, and the induction proceeds. (Without the t_*^k-term the Killing part of X would not be removed; this corrects an earlier version.)

□

Appendix A. Computer-assisted steps

Except in item C9, all computations use exact arithmetic over $\mathbb{Q}$ or $\mathbb{Q}(i)$. The code is available from the author; the script names are given in brackets.

C1. *Dual potential* [`step39_c1_symbolic.py`, `step89_c1_kappa.py`]. After Vieta elimination, the numerators of $n_1 - (\lambda C_0 + c_1)$ and $n_2 + 3$ are divisible by $s^2 - 4$ as polynomials.

C2. *Window margin* [`cert3/win_flint.py`, `cert3/margin_red.py`, `cert3/fullcert_m.py`, `cert3/nncert.py`]. With $\lambda = \lambda_{\mathrm{cr}}$, the quantity $(\Lambda/3)(L(z_0) - A/(2z_0) - 2t)$ is quadratic in m. After reducing a^2 modulo $b^2 - e_2 b + e_4$ and writing the sign condition as $F_1 > 0$ and $P = F_1^2(e_2^2 - 4e_4) - Y^2 > 0$, every irreducible factor (degrees up to $(34, 36, 8)$ in $(r_0, 1/r_2, \tau)$) has nonnegative, not identically zero, Bernstein coefficients on $[0,1]^3$. This proves strict positivity on the open cube.

C3. *One sign change* [`step134_P1new.py`, `step135g.py`, `nnd.py`]. The combinations g_1, g_2 for $k - 1 = 2, 3, 4$ and both cases $a_0 = 0$, $a_0 = A_{c2}$, factored and certified factor by factor by the same Bernstein criterion in two variables.

C4. *Seam inequality* [`step172.py`, `step173_build.py`, `cover4.py`]. In $(X, T, \sigma, \rho) \in (0,1)^4$: $\partial_A N < 0$ (one box); $N \geq 0 \Rightarrow Q > 0$ at $A = 0$ (51 boxes); the corresponding statement at $A = A_*$ (25 boxes); and both at $\rho = 1$ (one box each). On every box, one of the two polynomials has nonnegative Bernstein coefficients, with a positive coefficient on every face not contained in $\partial[0,1]^4$.

C5. *Zero frequency* [`step125_sigma0_check.py`]. The Teukolsky–Starobinsky identity at $K = 0$ for the $s = -2$ equation, and the 4×4 determinant. For $s = +2$ one substitutes $R_{+2} = \Delta_r^{-2}F$: then $\Delta_r^{-2}(\Delta_r^3 R'_{+2})' = \Delta_r^{-2}(\Delta_r F'' - \Delta_r' F' - 2\Delta_r'' F)$, with $\Delta_r'' = -4\Lambda r^2 + 2(1-\alpha)$. This matches the r^2-terms of the two potentials, $-10\Lambda r^2 + 8\Lambda r^2 = -2\Lambda r^2$, and the constants are absorbed by the relation between the spin ±2 angular eigenvalues at $\sigma = 0$.

C6. *Type D identities* [`step92_ghp_check.py`, `step93_*.py`, `step95_pd_xi.py`, `step96_zeta.py`, `step97_r5_pd_check.py`]. These are checks of identities that are already derived by hand; the derivations do not depend on them.

C7. *Carter ansatz* [`step102_carter_linear.py`]. The Ricci tensor of the ansatz with symbolic $\Delta_R(R)$ and $\Delta_I(I)$, and the solution of the resulting linear equations.

C8. *Schwarzschild–de Sitter Teukolsky–Starobinsky identities* [`step126_sds_ts.py`, `step127_thorn_ts.py`]. These are the two identities (6.2). The first is an operator identity. The second is checked by reducing modulo the radial equation; the remainder coefficient of R' vanishes identically, and the coefficient of R is the constant $\mathcal{C}_\ell$. That this coefficient is independent of r is also a consistency check on the radial potential used.

C9. *Cosmological side* (Proposition 3.9). Variables $(x, v, \tau) = (r_0, 1/r_2, \text{window position}) \in (0,1)^3$; $b = a^2$ enters only through its physical root b_+ of $b^2 - e_2 b + e_4$, which has no rational parametrisation (for fixed r_0 the curve is a smooth plane cubic). Signs at b_+ are

decided from the reduction $F_1 b + F_0$ via (F_1, Y, P) with $Y = -(2F_0 + F_1 e_2)$, $P = F_1^2 D - Y^2$, and, when that is ill-conditioned (large r_2), by interval evaluation of the unreduced polynomial on an interval enclosure of $b_+ = 2e_4 v^2/(e_2 v^2 - \sqrt{D v^4})$, which is smooth at $v = 0$. Each condition (i)–(v) of Step 6 is factored into odd-multiplicity factors (squares dropped); boxes are subdivided until every factor has a certified sign. The atlas consists of the cube and blow-up charts at the faces $x = 1$ (extremal), $v = 0$ (Kerr), $v = 1$ (Nariai/ultracold), $\tau = 1$ and their intersections (charts KA, KB, VT, XT, T1–T3, U1–U3, EA, EB, with second blow-ups where needed). *Which certificate proves what.* (i): $L(t)$ (with $\lambda_{\rm cr}$) at $m = 1, 2$ and its m^2-coefficient are certified positive wherever $2\delta \geq t - 1$ (so, by the concavity in Step 3, $L(t) > 0$ there for every m) [`smu_build.py`, `smu_cover7.py`, `smu_corner.py`]. (ii): the gate-free form $W_C := 2q_1^2\theta - 1.05\, t(t-1)(-L(t)) \geq 0$ in z_0-free form (Step 3), for $m = 1, 2$ and the m^2-coefficient, on the cube outside the corner square $\{r_0, 1/r_2 \geq 7/8\}$; inside it, (i) and $A \geq 0$ reduce (ii) to a quadratic inequality in τ, certified on charts EA, EB [`sca_build.py`, `gcover.py` (exact), `scg_tau.py`, `cover2d_f.py`]. (iii) [`za_build.py`] and (iv) [`zp_build.py`]: full atlas, with the closure "$\varpi_m \leq 0$" allowed on each box. (v): one box on the whole cube [`aux_build.py`, `cover3d_g.py`]. *Arithmetic.* The polynomial cores and the condition (i), S_μ, are handled in exact integer Bernstein arithmetic. For the remaining conditions the Bernstein coefficients are computed exactly in $\mathbb{Q}$, divided by their maximum modulus and rounded once to double precision. Subdivision uses de Casteljau midpoints. Alongside each array A we carry the array B obtained by applying the same subdivisions to $|A_{\rm root}|$. Each midpoint step adds an error of at most u times the average of the moduli of its inputs, with $u = 2^{-53}$. So after subdivisions whose degrees sum to Σ, every entry satisfies $|A_{\rm float} - A_{\rm exact}| \leq u(1+\Sigma)B$, up to a factor $1 + O(u\Sigma)$. A sign is accepted only if $A > c\,B$ (or $A < -c\,B$) entrywise, with $c = 10^{-13}(k+4)$ at depth k. Since every per-axis degree is at most 122, c exceeds $u(1+\Sigma)$ by a factor of at least 7. The root b_+ is enclosed with interval arithmetic (`mpmath.iv`) evaluated directly in chart coordinates. Scripts: `kds/cert3/{smu,sca,scg,za,zp,aux}_build.py`, `cover3d_h.py`, `cover3d_f.py`, `cover3d_g.py`, `cover2d_f.py`, `atlas.sh`.

The Bernstein enclosure routine was tested against random polynomials and random subdivisions [`cert2/test_bern.py`].

Statement on the use of AI

In preparing this work I used Claude Opus 5.5 (Anthropic) to develop and check proofs, write and run the symbolic and certificate code, search and check the literature, and draft and edit the text. The proofs were also checked in several adversarial review passes, each run by an AI model. These passes found errors, including invalid earlier versions of the large-A regime of the event lemma and of the passage to the upper half plane, and every error found was corrected. Numerical experiments were used only to guide the search for proofs, and no result depends on them. No human expert has yet checked the argument. I have reviewed and edited the content and take full responsibility for it.

Data availability

The code for the computer-assisted steps is available from the author on request; script names are given in Appendix A.

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